



MIZAN TEPI UNIVERSITY

SCHOOL OF COMPUTING AND INFORMATICS

DEPARTMENT OF INFORMATION TECHNOLOGY

Title: Yeki Wereda Court Information Management System

NO	Name	ID
1	ABNET GEREMEW	MTUUR0170/14
2	HAMID TAREKE	MTUUR1601/14
3	EBENEZER SYOUM	MTUUR1148/14
4	MIHRET FIKRU	MTUUR2269/14
5	ABREHAM AMANUEL	MTUUR0181/14

Advisor's Name: Mr.Amir Ali
Submission Date: 08/09/2017 E.C

Tepi, Ethiopia

Declaration

The Project is our own and has not been presented for a degree in any other university and all the sources of material used for the project have been duly acknowledged.

Name	Signature	Date
1. HAMID TAREKE	-----	-----
2.EBENEZER SEYOUM	-----	-----
3.ABINET GEREMEW	-----	-----
4.ABREHAM AMANUEL	-----	-----
5.MIHRET FIKRU	-----	-----

Advisor Approval Page

To: Information Technology department

Subject: Industrial Project Submission

This is to certify that the industrial project entitled yeki wereda Court information management system submitted in partial fulfillment of the requirements for the degree of bachelor of science in Information Technology, has been carried out by the group members under my supervision. Therefore, I recommend that the students has fulfilled the requirements and hence hereby they can submit the project to the department.

Advisor name

Signature

date

AMIR ALI

Examiner Approval Page

It is approved that this project has been written in compliance with the formatting rules laid down by the faculty

Examiner-1	Signature	Date
-----	-----	-----
Examiner-2		
-----	-----	-----
Examiner-3		
-----	-----	-----
Examiner-4		
-----	-----	-----
Examiner-5		
-----	-----	-----

Chairman name	Signature	Date
-----	-----	-----

Acknowledgment

This project is not done only by our effort. It has got an input from different individuals. First of all, the project team would like to greatly thanks to (Allah) God, who gave us the power to understand and for better analyzed the system. Secondly, Advisor Mr. Amir.A who gave sounding advice help us guide this project in the right direction, in which he enforce us to have a massive knowledge about the project to easily familiarize with the concept of project development. Thirdly, the project team would like to thank Tepi town court officers especially Mr.Zane Adelo for his full willingness of interview, interest in answering to project team numerous questions, telling how the existing system works for better understanding to the system.

Abstract

Now a days, using computers to simplify tedious manual work is significance. Because using computer gives many benefits like speed, accuracy, storage capacity, security, flexibility, cost reduction and minimizing tedious manual work. Tepi town court office was limited on manual works. This has a limitation for customers wasting large amount of time in front of the office to register and high consumption of resources.

To overcome this problem we are proposing the new web base system for Tepi town Court Information management System. Customer can find advocators by using online system. So that no needs waste the time and resource to find advocators.

This system is developed with step by step from planning, analysis, design, and then implementation. With the Unified Modeling Language (UML) diagram such as use case, activity diagram and sequence diagram is drawn. The system flow and the actor are drawn to understand how the system works does and who the actors that interact the system are.

The programming language used to develop this project is **PHP and HTML** .Database application is **MySQL**.

Table content	page
Acknowledgment	IV
Abstract	V
List of Tables	VIII
List of Figures	IX
List of Abbreviations and Acronyms	X
CHAPTER ONE	1
1 INTRODUCTION	1
1.1 Background of the Organization	1
1.2 Background of the Project	1
1.3 Statement of the problem	2
1.4 Objective of the project	3
1.4.1 General objective	3
1.4.2 Specific objectives	3
1.5 Methodology of our project	3
1.5.1 Data Sources	3
1.5.1.1 Primary Data Source	3
1.5.1.2 Secondary Data Source	3
1.5.2 Data gathering Techniques	4
1.5.3 Systems Analysis and Design	4
1.5.4 Overview of the Project	5
1.5.5 Development Tools and technology	6
1.5.5.1 Software Tools	6
Front end technology	6
Back-end technology	6
1.5.5.2 Hardware Tools	7
1.6 Scope of the project	8
1.7 Feasibility	8
1.7.1 Operational Feasibility	8
1.7.3 Behavioral /Political feasibility	9
1.7.4 Economic Feasibility	9
1.8 Limitation of the project	9
1.9 Significance of the project	10
1.10 Task and Schedule	11
1.11 Cost	11
1.11 Cost analysis of the project	12
1.12 Communication plan	12
1.13 Project Team Composition	13
CHAPTER TWO	14
2. EXISTING SYSTEM	14
2.1 Description of Existing System	14
2.2 Problems of the Existing system	14
2.2.1 Performance	14
2.2.2 Security and Controls	14
2.2.3 Economic	15
2.2.4 Data storage problem	15

2.2.5 Efficiency	15
2.3 Actors in the Existing system	15
2.4 Business rule of the system	15
2.5 Report generated in the existing manual system	16
2.7 Alternative options to address the problems of existing system	16
CHAPTER THREE	17
3. PROPOSED SYSTEM	17
3.1 overview of the proposed system	17
3.2 Functional Requirement	17
3.3 Non Functional Requirement	18
CHAPTER FOURE	19
4. SYSTEM DESIGN	19
4.1 System Design Using Object Oriented Paradigm	19
4.1.1 Design Goal	19
4.2 Use case diagram	20
4.2.1 Actor and use case identification	20
4.3 use case diagram	22
4.3.1 Use Case Description	23
4.4 Sequence diagram	33
4.5 Activity diagrams	39
4.7 Class diagram	49
4.9 Persistent modeling/database design	50
4.10 Component Diagram	53
4.11 Deployment Diagram	54
CHAPTER FIVE	55
SYSTEM IMPLEMENTATION	55
5.1 Introduction	55
5.2 Sample code	56
5.3 User interface	60
5.4 Testing	62
5.5 User manual preparation	62
5.6 Training	62
5.7 Start-up strategy	62
5.8 Conclusion and recommendation	63
5.8.1 Conclusion	63
5.8.2 Recommendation	63
Reference	64
Appendix	65

List of Tables

Table 1: Software tools.....	8
Table 2 : hardware tools.....	8
Table 3: Task and Schedule.....	12
Table 4: cost of the project.....	13
Table 5: communication plan.....	13
Table 6: project team composition.....	14
Table 7 :Actor and use case identification.....	21
Table8 :Use Case Description for Login.....	23
Table :9 Use case Description for new case registration.....	24
Table 10: Use Case Description for Create account.....	25
Table 11: Use Case Description for update account.....	26
Table12 : Use case Description for Give appointment.....	27
Table13 : Use case Description for case registration.....	28
Table 14 : Use Case Description for View Appointment.....	29
Table 15 : Use case Description for generate report.....	29
Table16 : Use case Description for view assigned case.....	30
Table 17: Use case Description for record decision.....	31
Table 18: Use case Description for view information.....	31
Table 19: Use Case Description for Give Feed back.....	32
Table20 : Use Case Description for view Feed back.....	32

List of Figures

Figure 1: Sequence diagram for login.....	32
Figure 2: Sequence diagram for give feedback	33
Figure 3: Sequence diagram for new case registration	34
Figure 4: Sequence diagram for create account	35
Figure 5: Sequence diagram for record decision	36
Figure 6: Sequence diagram for view information	37
Figure 7: Activity Diagram for Login	38
Figure 8: Activity Diagram for create account	39
Figure 10: Activity Diagram for record decision form	40
Figure 11: Activity Diagram for appointment page	41
Figure 12: Activity Diagram of officer home page	42
Figure 13: Activity Diagram for search case information	43
Figure 14: Activity Diagram for view appointment	44
Figure 15: Activity Diagram for feedback	45
Figure 17: Activity Diagram for view decision	46
Figure 18: state chart New case registration	47
Figure 19: state chart login page	47
Figer 20:state chart record desion	48
Figure 21: Class diagram	49
Figure 22 : Persistent modeling	50
Figure 23 :component diagram	53
Figure 24 : deployment diagram	54

List of Abbreviations and Acronyms

BR -----	Business Rule
MTU-----	Mizan Tepi University
MS.....	Microsoft
OOA.....	Object oriented analysis
OOD.....	Object oriented design
PHP.....	Hypertext Preprocessor
SQL-----	Structural Query language
PC.....	Personal computer
RAM.....	Random accesses memory
UML-----	Unified model language
UC-----	Use case
TTCIMS-----	Tepi Town Court Information Management System
DB-----	Database
Pk-----	Primary key
Fk-----	Foreign key
UI-----	User Interface
OOSD-----	Object oriented System Design
HTML-----	hypertext markup language
CSS-----	cascading stylesheet

CHAPTER ONE

1 INTRODUCTION

1.1 Background of the Organization

The court office was established during the "Derg" regime. Before this time disagreement and criminal cases were solved in traditional way. Starting from that time it passes many obstacles to reach today's position. The court office of tepi town yeki wereda established in 1996 E.C started to serve the customer in modern ways. The organization is located around the Tepi town administration on the right side and on the left side of old police station in hibret kebele 02. The purpose of the court is to provide legal decision for those plaintiffs and defendants based on their evidences as well as for those who commit crimes. The lawyers make decision based on the constitution of the country.

The court information management system addresses two major cases those are civil case and criminal cases. It has the responsibility to support, manage, giving response to the case, taking case from the kebele (mahiberawifirdeshengo) to wereda level court office.

We have seen some problems in data processing and handling. During the process there are many difficulties of manual processing of files/data. Not only this but also giving comments for one court is difficult because of distance.

Now we are going to develop software that can solve problems of data processing, data handling and etc.

1 .2 Background of the Project

The system based on the court information management system that improves the manual data processing. Court Information Management System is document management system or handling of data, data flow system, appointment and also concern with a customers' comment. There are two major types of cases. Those are civil law and criminal law. Each of them contains their own sub branches. In order to execute those cases some process takes place. As we visited the work place of how the court information management system takes place. we have seen some problems in data processing in tepi town and handling. During the process there may be many difficulties of manual processing of files/data.

Not only this but also giving comment for one court may be difficult because of distance. This means there is no online giving comment service before for the court found on the distance. Now we are going to develop software that can solve problems of data processing, data handling, give appointment, assign case, registering case and view assigned case etc. we can achieve our aim by using hardware (pc) and software (programming language). The main purpose of doing this project is that to save person's file from damage, store files forever, secure files that must be secured, reduce costs and time. There will be data communication from one office to the other. There will be many advantages after the end of our project.

1.3 Statement of the problem

Court information management system of tepi town yeki wereda does on both criminal and civil case type. As mentioned above these major types contain so many sub branches. Criminal case is initiated when crime is committed, then investigation held to collect information and evidence about what is happened to determine who committed the crime. After that the accused is charged with the crime and taken into custody. Based on evidence, then the judge determine whether he/she guilty or not guilty. At the end the decision is presented. This process is done manually.

We have seen the following problems as follows

Difficulty of storing the documents neatly

Files may be stolen by thieves, robbers or internal attackers.

Files may be fade because of long life

Files may be destroyed by natural disasters like fire, flood

Customer can't see some news on distance.

Difficulty of data store in secure way.

Customer can't inform appointments day and time remotely or from every where.

Recording examination, cross examination and reexamination may difficult while collecting evidence from witness.

Difficult to giving appointments for customers.

Difficult to evaluate every judge decisions.

1.4 Objective of the project

1.4.1 General objective

Our main objective is to develop web based court information management system for tepi town yeki wereda .

1.4.2 Specific objectives

The target of this work is to design and implement court information management system. The following are the specific objectives.

To allocate judges to cases.

To store judiciary case files electronically in databank

To develop a system that will provide an interface to easily capture court case information.

To guide and support the easy updating and retrieving of judiciary case files

To schedule appointment date.

To Implement and Test new system by using different testing methodology.

To Deploying the tested system.

1.5 Methodology of our project

1.5.1 Data Sources

Both primary and secondary sources of data are used to gather information.

1.5.1.1 Primary Data Source

By interviewing respective personal of the workers on the court office.

Collect end users feedback about the existing system from the customers and judges.

1.5.1.2 Secondary Data Source

Relevant documents and techniques that helps to improve the performance of website

Just after data collection requirement analysis is done and the next stage of the project is system design, and object oriented design is used to design the system so that the output will enable to visualize the objects in the system in a simple and understandable form.

1.5.2 Data gathering Techniques

Since data gathering is the process of gathering and measuring information on different area of the system.

We will implement the gathering technique in the following ways :

We observe how the site works.

We observe how the site really needed to be managed and modified.

We observe how data 's are not updated frequently on the site.

1.5.3 Systems Analysis and Design

It is the process of breakdown an entire system into module, analyzing each module separately and determining the relationship between the system development process we used system modeling.

Requirements Gathering and Modeling

Identify actors (users or other systems) and their interactions with the system.

Create use case diagrams to describe system functionality from the user's perspective.

Object Identification and Modeling

Identify key objects/entities in the problem domain (e.g., Case, Judge, Customer).

Define their attributes and methods/operations.

Create class diagrams to show relationships (inheritance, association).

Behavioral Modeling

Model the interactions between objects using:

Sequence diagrams – show the flow of messages.

Activity diagrams – represent workflows and logic.

State diagrams – show how objects behave in different states.

System Architecture Design

Define components and how they interact (component diagrams).

Design the deployment structure – where software will run (deployment diagram).

Data Persistence

Model and design the database using persistent modeling.

Ensure that object data can be saved and retrieved from storage (like MySQL in your project).

User Interface Design

Design intuitive and accessible interfaces based on user roles (admin, judge, customer).

Use technologies like HTML/CSS/JS for front-end and PHP/MySQL for back-end (as in your project).

Security and Access Control

Implement role-based access (e.g., only a judge can record a decision).

Secure data using login credentials, encryption, and validation.

System Testing and Validation

Verify that each component works individually (unit testing) and together (integration/system testing). Validate that system fulfills all specified requirements.

1.5.4 Overview of the Project

The new system will include solutions for the existing system problems by giving services better than the older one as much as possible. After observing the current manual court information management system and evaluating all the problems that occurred during every activity on the existing system, the project team has desired to design a web-based court information management system for the tepi Town Yeki Wereda with the following improvements:-

Performance:- the system will have a faster response time and use minimal space usage.

Speed:- regarding the speed the system will generate output within a short time.

Reliability:- the system is reliable by analyzing each business rule set by the organization

Security:- the system provides or contains a user name and password for each user based on their privilege this increases the security of workflow.

For administrative it is best to see the weekly, monthly, and yearly reports and activities. It also helps to know how the judge is resolving problems and it helps to get feedback from the clients about the court information

management system. For the organization, the system will help save cost and time.

The new proposed system is used to minimize the number of tasks labeled to the employees.

Economy:- when we apply a Web-based court information management system there will be areas in which cost will be reduced. Example as a result of a new system the payment too many employees will be reduced. Reduce the cost of paper and other materials.

Efficiency:- the Web-based court information management System by itself is short and clear and in this system there is no duplication of data through the new system is powerful to manage things.

1.5.5 Development Tools and technology

1.5.5.1 Software Tools

The following requirement software tools are used to implement our project work.

Front end technology

HTML: used for user interface design

CSS: used for styling HTML user interface.

JavaScript: Makes it interactive

Bootstrap: reponsive design mobile friendly site

Browser: Google chrome , Mozilla Firefox

Back-end technology

Server: Apache HTTP Server is used run the web server within Windows OS.

Database: MySQL is common database in creating dynamic websites it is a high- speed database.

PHP: is a scripting language that can be used to access data from the database.

Generally, Apache, MySQL, and PHP are open-source components that can be installed individually, they are usually installed together.

NO	Activity	Tools / programs
1	HTML, CSS, JS	Client-side programming
5	PHP	Server-side scripting
6	MYSQL	For database
2	MS windows	Platform
3	Microsoft word	to prepare documentation
4	XAMPP	For database
7	MS Power point	For presentation
8	Draw.io	to draw UML diagram
9	Browser	Google chrome , Mozilla Firefox

Table 1: Software tools

1.5.5.2 Hardware Tools

NO	Hardware name	Description
1	Printer	For printing out design prototype and document in hard copy
2	Computer	To do computing process
3	Internal/external hard disk	For data storage and backup
4	LCD projector	For presentation

Table 2 : hardware tools

1.6 Scope of the project

The scope of our project is to implement a web-based court information management system, which is aimed to avoid the problems associated with the manual system which focuses on tepi town.

Accurate way of recording and storing information into the database.

Administrator can control everything and assign roles to each of them as they see fit.

Make the system high secure, efficient and effective

View assigned cases that case distributor are assign to Judges.

The assigned judges will set decide judgement day.

To be able to notify customers of the judgement day by message when the judges assigned.

Register any customer that wants to open new case.

President and vice president view reports.

avoiding the data redundancy of the system.

Search the files in short times from many files.

Update employee's information.

Accept comment that customers are writing for court offices.

1.7 Feasibility

1.7.1 Operational Feasibility

The system requires very little human power. It provides a simple user interface in which the employee of the office can access easily, and more accurately. Maintenance is less and easy. The system will have GUI and very little user training is required to learn it. The system is so simple to use, so no special skills will be required to use the system. So the system is operationally feasible.

1.7.2 Technical Feasibility

The system is to be developed by using technologically system development techniques such as PHP, Java script, CSS, and MySQL database without any problems and the group members have enough capability to develop the project. So the system will be technically feasible.

1.7.3 Behavioral /Political feasibility

The political feasibility is a feasibility that makes sure the system does not break law. Our system does not violent with any law, hence the system is politically feasible.

1.7.4 Economic Feasibility

The system to be developed is economically feasible and the benefit is out weighing the cost. Since this project already computerizes the existing system, now the reduction of cost for materials used in manual operation becomes beneficial to the organization.

1.8 Limitation of the project

The system does not send appeals to the high court and Supreme Court because there is no interconnected web-based system with zonal court office. Our system has no online judgment

Our system has no sound and video record.

1.9 Significance of the project

After the implementation of this project, it provides various benefits. as follow

Can save their financial budget.

Customers will give their comment online without going there.

Customer informed appointments day and time remotely.

Adequate and fast communication with judge and other coworker.

Assigning new case to judge easy and secure more with this this system.

The judges find appointment date and time easily for many case in short period of time.

Recording examination, cross-examination, and reexamination is easy while collecting.

It reduces the wastage of time during searching and modifying employee

Reduce cost paid for extra hardware materials like pen, paper, and printing.

Decrease Error in filing the client document.

Information about court can be accessed any time from any place based on given privilege.

1.10 Task and Schedule

NO	Activity	Start	Finish	Nov 17	Dec 17	Jan 17	Feb 17	Mar 17	Apr 17	Mar 17	Jun 17	
1	Proposal	25/03/17	02/04/17									
2	System analysis	03/04/17	04/05/17									
3	System Design	06/05/17	01/06/17									
4	Implementation	02/06/17	03/09/17									
5	Testing	04/09/17	02/10/17									

Table 3: Task and Schedul

1:11 Cost analysis of the project

NO	Material Required	Amount	Price per unit	Total prices
1	Flash	3	350	1,050 birr
2	Personal computer	4	35,000	140,000 birr
3	Note book	3	120	360 birr
4	pen	5	30	150 birr
5	paper	1	500	500 birr
Total				142,060 birr

Table 4: cost of the project

1.12 Communication plan

Communication plan describes the schedule of group member or the time of communication plan to develop project. The group member meets four days in each week.

day	Time	Place (class room)
Monday	Local(4:00-6:00)pm	LAB
Tuesday	Local(3:00-4:00)	LAB
Thursday	Local(3:00-5:20)pm	LAB
Saturday	Local(2:00-5:00)am	LAB

Table 5: communication plan

1.13 Project Team Composition

NO	NANE	IDNUMBER	Responsibility
1	HAMID TAREKE	MTUUR/1601/14	-documentation -Requirement analysis -Data gathering
2	EBENEZER SYUM	MTUUR/1148/14	-Implementation -Data gathering -Designing -group Coordinator
3	ABNET GEREMEW	MTUUR/0170/14	-Data gathering -Requirement analysis -Data gathering
4	ABREHAM AMANUEL	MTUUR/08/14	-System Analyst -Data gathering -Testing
5	MIHRET FIKRU	MTUUR/2256/14	-System Analyst -Data gathering

Table 6: project team composition

CHAPTER TWO

2. EXISTING SYSTEM

2.1 Description of Existing System

The existing court information management system of Tepi town court provides services to society. It handles two types of cases, civil case and criminal case. Civil case is proceeding between two individuals or organizations. This type of case is family issue, property and succession. In a civil case, the accuser takes the case to the court. After the court opens the case, the case will be verified. The case may be accepted or rejected. The case are rejected, if written words should not be net and clear, there should not have formalize stamp on the evidences that comes from police Station to open a new case. If the case is accepted, the law officer assign case to judge. Then, plaintiff detail and the gathered information will be recorded. After this process, the judge gives appointment to the accuser. Then, the law officer writes summon for the accused. Accuser sends the written summon to the accused person. On the day of the appointment, the law officer collects the information and gives it to the judge. The evidence and other information are recorded on paper. This process is repeated in this manner starting from hearing to rendering decision. The process of Criminal case is the same as civil case. What makes it different is that it can be conducted between state and individuals. In both civil and criminal cases, if the case is in progress, its files are called active files. After Some process, the case will be given a decision. The judge will record the final decision and the files of the case are archived files. The whole process of case handling takes place manually. That means, from opening of the case until it gets a final decision.

2.2 Problems of the Existing system

2.2.1 Performance

The performance of the existing system does not provide fast response. Because, it is difficult to access data from the stored catalogue and also it is time wastage and energy consuming.

2.2.2 Security and Controls

Every record of the existing court information management System is stored in the paper based.so, it is difficult to control and secure these manual records, since it doesn't have any authentication and authorization system.

2.2.3 Economic

Due to the operation that is done by the hand most of the activities are causes to high consumption of resources like papers, man power, time etc. This makes the existing system costs are too high.

2.2.4 Data storage problem

Data are not easily accessible due to place in different location. Difficult to change and edit. Data redundancy that leads to inconsistency.

2.2.5 Efficiency

Because of poor communication between court offices there is no efficiency, there is delay of activities. They may not get notification of case appointment date. The document collected for the case is not documented in organized form so it is difficult to search and give decision, as a result the efficiency of the decision is late.

2.3 Actors in the Existing system

This illustrates the actors involved in the current system. Those actors in the current manual system are: -

- Administrator
- Law Officer
- Case distributor
- Judge
- Customer
- Presidant

2.4 Business rule of the system

There are some business rules and constraint to prevent any violation during the process. This focuses on access control issue[7].

BR1. There should be enough evidence to open a new case.

BR2. Witness and accused should be above 18 years old and witness should be two or three.

BR3. The written words should be net and clear.

BR4. There should be opened file date on the new case.

BR5. There should be case number on the new case.

BR6. The law officer should be sign on the case paper to lead it to the judge.

BR7. All the forms should be filled correctly and there should be not be blank.

BR8. The customer must come to the office on the appointment day.

BR9. Any appointed case must be seen within 7 up to 10 days

BR10. The judge doesn't have to be related to the accuser or accused person.

BR11. Any accused or accuser who can't defend by himself can represent advocator.

2.5 Report generated in the existing manual system

There are many reports that reported through the process of the court information management system there is a report that goes to the law officer that holds information about the case and evidences that help to make sure to open a new case, and also there is a report that goes to the judges office to choose which is the case, the judge see last there is a report that goes to the accused or witness to come over to the court. Reports are generated in every six months to evaluate the performance of the employee.

2.7 Alternative options to address the problems of existing system

After we identified the real problem of the existing system which is in a manual system, the team suggests an alternative option to overcome the problem. These alternative options are:-

Changing the manual system into web based system.

Changing the manual system into Expert system.

Changing the manual system into Mobile application.

Now a days, website need to better than before. Because people starting to get used to browsing website are user-friendly, fast, secure, and can view on other device that they can carry with such as smartphone. so all stakeholders the system need to be Responsive. Due to this the team has analyzed all of the alternative options based on the ability of performance, information flow and service to the users and efficiency. This analysis has enforced to select the web based system.

CHAPTER THREE

3. PROPOSED SYSTEM

3.1 overview of the proposed system

After observing the current manual court information management system and evaluating all the problems occurred during every activity on the existing system the team has desired to design a web based court information management system for Tepi town yeki wereda court. The proposed system of our project changes the existing manual process of the organization to new system.

To eliminate drawback of the existing system, we develop web based system. The new system gives full system functionality that is needed by system user to perform system functionality. The proposed system is: Store data into the database so, it can be easily received and used at any time. Provide highly security to data. Very easy to use. Suitable for employees find customers' information in a short time. Suitable for register for customer information and record the final decision of customer's case. Used to store more data in a small space. Used for customers easily find the advocators by searching in the web based court information management system. The proposed system will use the major functionality of the existing system able to advance accordance with speed of performance, security and reliability of the system, by using different object oriented techniques in order to overcome the current problems.

3.2 Functional Requirement

Functional requirements are the functions of the proposed system. It describes the interactions between the system and its environment independent of its implementation. The major functionalities of the system are the following:

User management

Adding new user account, modifying recent users account.

User can be register and login

Different access level (eg Admin, judge, case distributer, low officer)

Case management

Create, update and delete case records

Case tacking status (panding, open, distribute and closed)

Case assignment

Document management

Update, store and retrieve documentation

Schedul hearing and calander

Schedule hearing and appointment

Notification and the upcoming events

Integration with calendar system

Reporting and analytics

Generate reports on case stastics

3.3 Non Functional Requirement

Nonfunctional requirements are requirements that do not directly affect the performance of the system. These requirements relate to the technical aspects of the system that has to be addressed, such as the security issue. Because nonfunctional requirements are requirements that are constraints on the system.

Some of the nonfunctional requirements are: -

Maintainability: The system will be developed for easy maintenance and future expansions. The system has to be easily understandable for the users and have to be easy for maintenance.

Security: The system has to be well protected unauthorized access. Encrypted Password. Administrator has more rights than the sub user.

Availability: The system should have to be functional at any given time.

Reliability: The proposed system will minimize crash during its runtime, since more than one user could use the system simultaneously. Good validations for user inputs will be done.

Avoid incorrect storage of records.

Performance: The system should have to perform in a proper way without any problem. Performance requirements define acceptable response times for system functionality.

User interface: the system should have to provide interactive and easily manageable user interface for the users.

Portability: This Website opens in any personal computer supporting windows, Linux, UNIX operating system.

Flexibility: The system keeps on updating the data according to the operations that takes place

CHAPTER FOURE

4. SYSTEM DESIGN

4.1 System Design Using Object Oriented Paradigm

System design is the transformation of the analysis model into a design model. System design can reduce gap between problem and an existing machine, decomposes the overall system into manageable parts and uses the principles of cohesion and coherence. Under system design court information management architecture, component, modules, interfaces and persistent data are mentioned in order to achieve specified requirements. Court information management system works in a networked environment and is going to be a web based application. Users of the system can access it from a local area network or from anywhere over the Internet. Therefore, the system design will use standard web based system development tools and technologies in order to allow users access the system seamlessly over a network.

4.1.1 Design Goal

The design goals of our system are inferred from nonfunctional requirements and explained as follows. It describes the qualities of the system that the developers should consider.

While developing Court Management System, there are a lot of design goals to be taken. The design goals are based on standard software design methodologies

User friendly: our system should be designed in easy way. It is not difficult to learn or to understand. The system interface will be well-organized, easy to locate different buttons and operations.

Performance: our system will be designed by considering performs tasks within a time frame that are acceptable for the user, and does not require too much memory.

Minimize cost: our system minimize the cost that perform some task.

Portability: our system is portable because it can work in different environments. This means, it is designed to run on different computers with compatible operating system and processes.

4.2 Use case diagram

4.2.1 Actor and use case identification

Use case diagrams are an UML diagrams used for capturing functional requirements of the system.

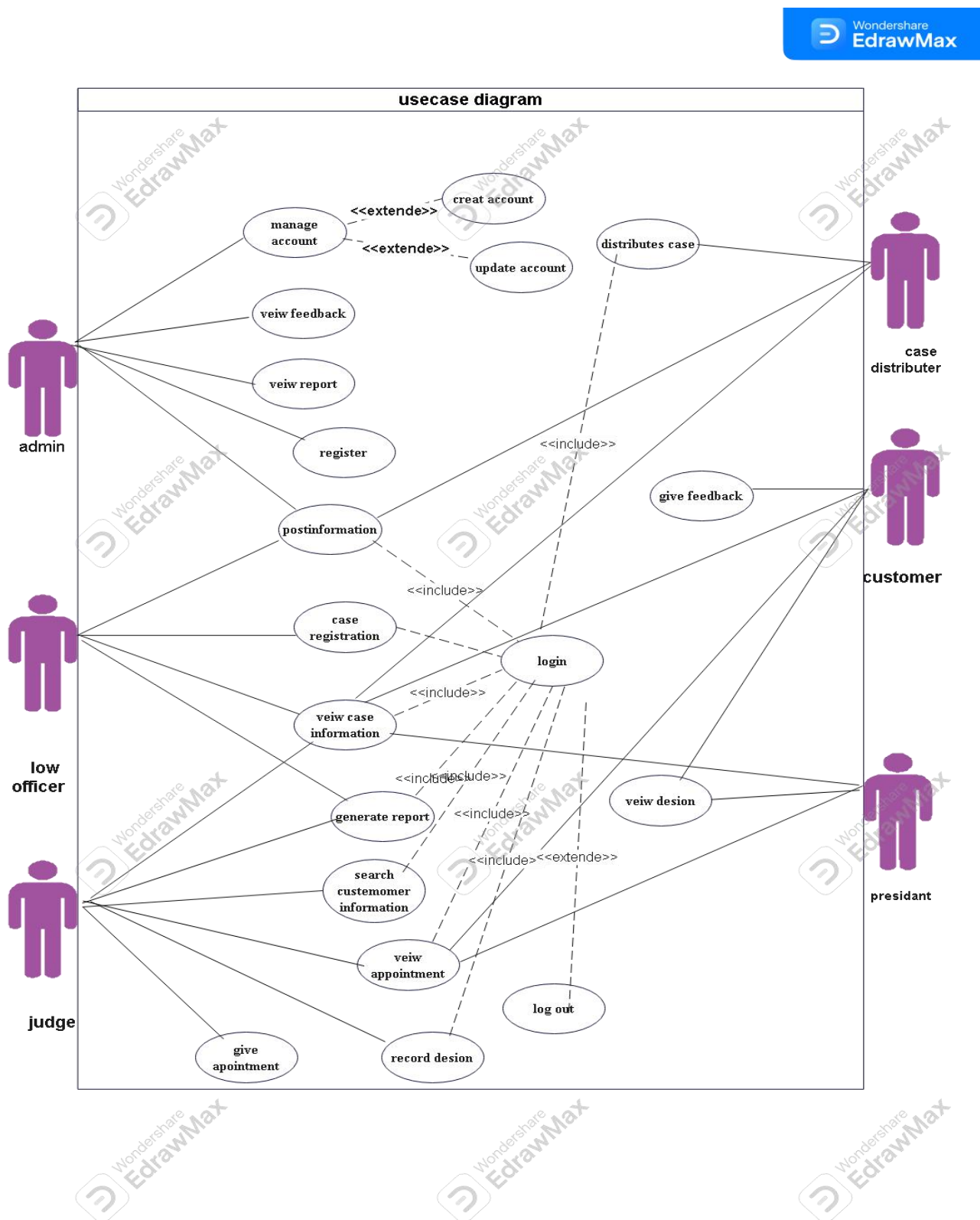
Each use case diagram describes a behaviorally related sequence of transaction in a dialogue between the user and the system. A use case diagram is a collection of actors, use cases, and their communications. Each use case is a complete course of events in the system from user's respective. It describes a sequence of action that provides a measurable value to an actor and draw as a horizontal ellipse[1]. An actor is a person, organization, or external system that plays a role in one or more interactions with the system and draw as stick man figure.[2]

Actors	Actors role(Attributs)
Administrator	➤ Mange account - Create account - Update account ➤ View Report ➤ View feedback ➤ Register judge,case distributer and low officer
Law officer	➤ Register case ➤ Generate report ➤ View case information ➤ Post information ➤ Search Customer Information
Case distributer	➤ View case information ➤ Post information ➤ Case distribut

Judge	❖ Judge process ➤ Record decision ➤ Give appointment ➤ View appointment ➤ View case ➤ Generate report
Customer	➤ View information ➤ Give Feedback ➤ View appointment ➤ View Decision
president	➤ View information ➤ View appointment ➤ View Decision

Table 7 :Actor and use case identification

4.3 use case diagram



4.3.1 Use Case Description

Use case number	UC1
Use case name	login
Actor/s	Adminstrator
Description	It allows user to login into the system
precondition	The users must have user name and password.
postcondition	The user able to access his/her privileges.
Basic course of action:	Step 1. The user click login menu. Step 2.The system displays the form. Step3.The user fills his/her user name and password and click login button. Step4. The system verifies the username and password. Step5. The system displays the appropriate page. Step6. The use case ends.
Alternate course of action	A4 Incorrect user name and password. 4.1 The system display reenter user name and password. 4.2 Go to step 3 to fill again.

Table8 :Use Case Description for Login

Use case number	UC2
Use case name	New Case Registration
Actor/s	Low officer
Description	It allows the law officer to register new case.
precondition	Law Officer must login in to the registration form.
postcondition	New case will be Successfully registered.
Basic course of action:	Step 1. User will login using UC1. Step 2. Open the new case registration link from menu. Step3. The system Displays the Registration form. Step4. The user fills all the information of new case and click Register button. Step5. The system verifies the filled information. Step6. The system displays successfully registered. Step7. The use case ends
Alternate course of action	A5. Incorrect information entry. 5.1. The system displays re_enter the information. 5.2. Go to step 4 to fill again.

Table :9 Use case Description for new case registration

Use case number	UC3
Use case name	Create Account
Actor/s	Adiminstrator
Description	It allows administrator to create for new user accounts.
precondition	He/she must login in to the create account form.
postcondition	The system successfully created account.
Basic course of action:	Step 1. User will login using UC1. Step 2. Open the manage account form. Step 3. The system Displays the Manage account form. Step 4. The administrator fills the information to create account and click create button. Step5.The system verifies the filled information. Step 6. The system display successfully created. Step 7. The use case end.
Alternate course of action	A5. Incorrect information entry. 5.1 The system displays reenter the information. 5.2 Go to step4 and fill again..

Table 10: Use Case Description for Create account

Use case number	UC4
Use case name	Update Account
Actor/s	Adiminstrator
Description	It allows administrator to update user.
Precondition	He/she must login in to the update form.
Postcondition	The system display successfully updated.
Basic course of action:	Step 1. User will login using UC1. Step 2. Open the manage form. Step3. The system Displays the Manage account form. Step4. The administrator. Fill the information that want to update and click button. Step5.The system verifies the filled information. Step6. The system display successfully updated message. Step 7. The use case end. Alternate course of action:
Alternate course of action	A5. Incorrect information entry. 5.1 The system displays reenter the information. 5.2. Go to step4 and fill again.

Table 11: Use Case Description for update account

Use case number	UC5
Use case name	Give Appointment
Actor/s	Judge
Description	It allows judge to give appointments for customers
precondition	He/she must login in to the home page.
postcondition	Judge successfully gives appointment.
Basic course of action:	Step 1. User will login using UC1. Step 2. Open the give appointment page from menu. Step3. The system displays the give appointment form. Step 4. The Judge fills the form. Step5.The system verifies the filled information. Step6.The system displays successfully appointment is given. Step7. The use case ends.
Alternate course of action	A5. Incorrect information entry. 5.1 The system displays reenter the form. 5.2. Go to step4 and fill again.

Table12 : Use case Description for Give appointment

Use case number	UC6
Use case name	Register case
Actor/s	Low Officer.
Description	It allows the law officer to register cases.
precondition	He or she must login in to the home page login form.
postcondition	Successfully registered.
Basic course of action:	Step 1. User will login using UC1. Step 2. Open the low officer page from menu. Step3. The system Displays the Registration form. Step4. The law Officer Fill the case information and click save button. Step5.The system verifies the filled information. Step6. The system displays registered successfully. Step7. The use case ends.
Alternate course of action	A5. Incorrect information entry. 5.1. The system displays re_enter the information. 5.2 Go to step 4 to fill again.

Table13 : Use case Description for case registration

Use case number	UC7
Use case name	View Appointment
Actor/s	Judge, customer,presidant
Description	It allows to view appointments.
precondition	He/she must login in to the View Appointment form.
postcondition	Successfully view selected information
Basic course of action:	Step 1. User will login using UC1. Step 2. Open the view appointment form from menu. Step3. The system Displays the view appointment form.

Table 14 : Use Case Description for View Appointment

Use case number	UC8
Use case name	Generate Report
Actor/s	Low Officer and Judge.
Description	It allows generating report in the database.
precondition	He/she must login in to the generate report form
postcondition	Successfully generate report as him/her wanted.
Basic course of action:	Step1. User will login using UC1. Step 2. Open the generate report form from menu. Step3. The system Displays the form. Step4. Select he/she want to generate a report. Step5. The system displays the selected report. Step 6. The user send report. Step7.The system verifies the send report. Step 8. The system display successfully generated. Step9. The use case ends
Alternate course of action	A7 .the system display error message. 7.1 The system display reselect. 7.2 Go to step 4 to fill again.

Table 15 : Use case Description for generate report

Use case number	UC9
Use case name	View Case information
Actor/s	Judge, Low officer,president
Description	t allows the Judge, Low officer, president to view the case.
precondition	He/she must login in to the home pag
postcondition	See the case.
Basic course of action:	Step1. The user will login using UC1. Step 2. Open view case form from menu. Step3. The system Displays the View case form. Step4. The system displays case or no case. Step5. The use case ends

Table16 : Use case Description for view assigned case

Use case number	UC10
Use case name	Record decision
Actor/s	Judge
Description	It allows the Judge record the decision.
precondition	He/she must login in to the home page.
postcondition	Successfully record the decision.
Basic course of action:	Step1. The user will login using UC1. Step 2. Open the record form from menu. Step3. The system Displays the Record Decision form. Step4. The judge record the decision in the record form and click Save button. Step5. The system display record successfully. Step6. The use case ends

Table 17: Use case Description for record decision

Use case number	UC11
Use case name	View Information
Actor/s	Customer
Description	It allows customer to view information that is the information of courts
precondition	Open view information form.
postcondition	He/she see information.
Basic course of action:	Step 1. Open the view information form. Step3. Select one from lists link. Step2. The system Displays the View information form. Step4. The system display information. Step6. The use case ends.
Alternate course of action	A1. Unknown the website

Table 18: Use case Description for view information

Use case number	UC12
Use case name	Give Feedback
Actor/s	Customers.
Description	It allows customer to give Feedback about court office.
precondition	The customer must know court office website.
postcondition	The customer give feedback successfully
Basic course of action:	Step 1. Open the give feedback form. Step 3. Customer write feedback and click send button. Step2. The system Displays the give feedback form. Step4.The system Displays successful. Step5. The use case ends.

Table 19: Use Case Description for Give Feed back

Use case number	UC13
Use case name	View Feedback
Actor/s	Administrator
Description	It allows administrator and law officer view any feedback.
precondition	He/she must login in to the home page.
postcondition	See Feedback.
Basic course of action:	Step1. The user will login using UC1. Step 2. Open view Feedback form from menu. Step3. The system Displays the View feedback form. Step4. The system displays feedback or no feedback. Step5. The use case ends.

Table20 : Use Case Description for view Feed back

4.4 Sequence diagram

A sequence diagram in a unified modeling language (UML) is a kind of interaction diagram that shows how processes operate with one another and in what order. It is a construct of a Message Sequence Chart. A sequence diagram shows object interactions arranged in time sequence

Basic Course of Action

1. The user wants to open and Login in to the system
2. The system Displays login page
3. The user fill his/her user name and password.
4. The system verifies username and password.
5. The system displays appropriate page
6. The use case ends

Alternative course of action

- A3. The user name /passwords invalid
- 3.1 The System displays error message.
 - 3.2 The system continues as step 3 to fill user name and password again

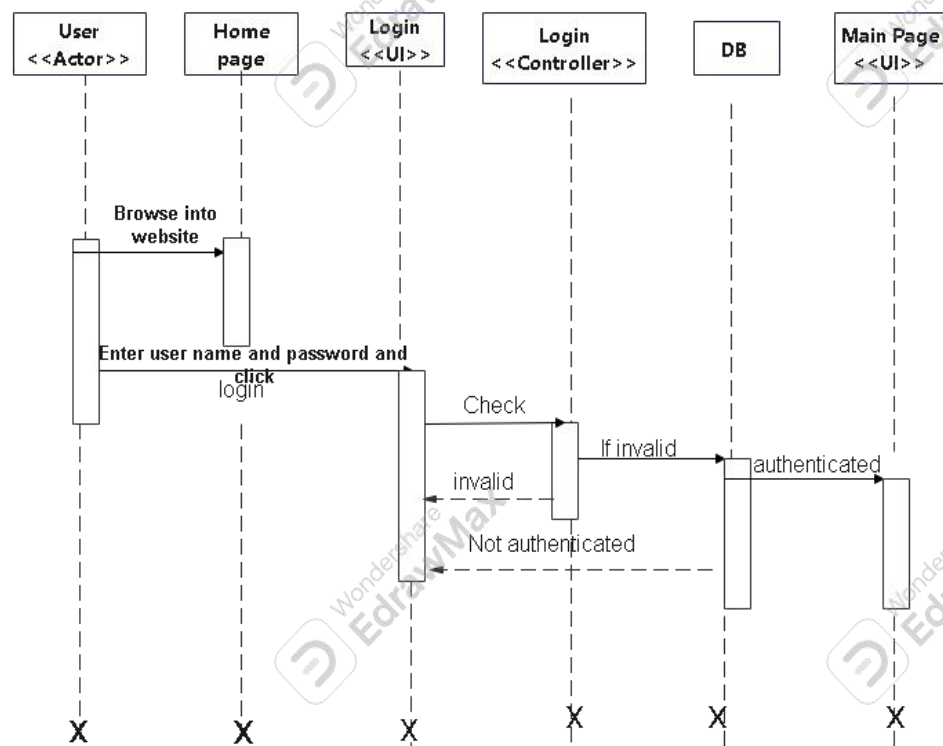


Figure 1: Sequence diagram for home page

Basic Course of Action

1. Open the give feedback page.
2. The system displays the give comment page.
3. The customer write comment on the form and click send.
4. The system displays successfully send
5. The use case ends.

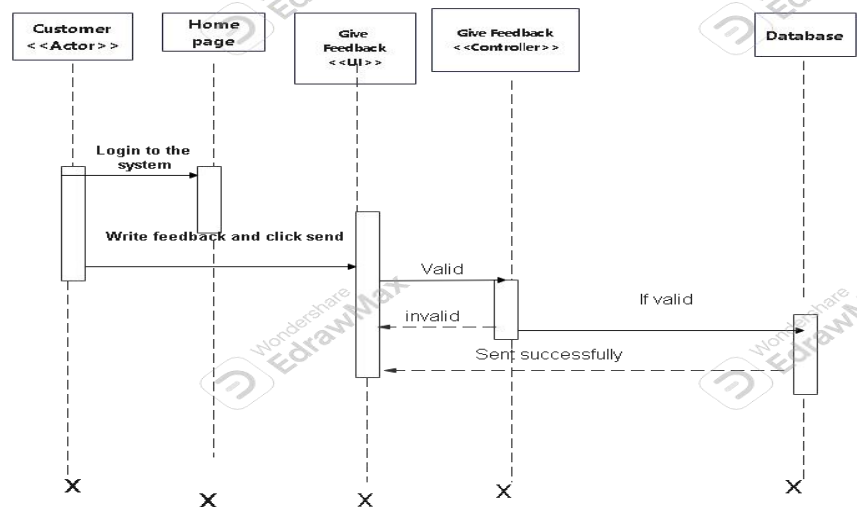


Figure 2: Sequence diagram for give feedback

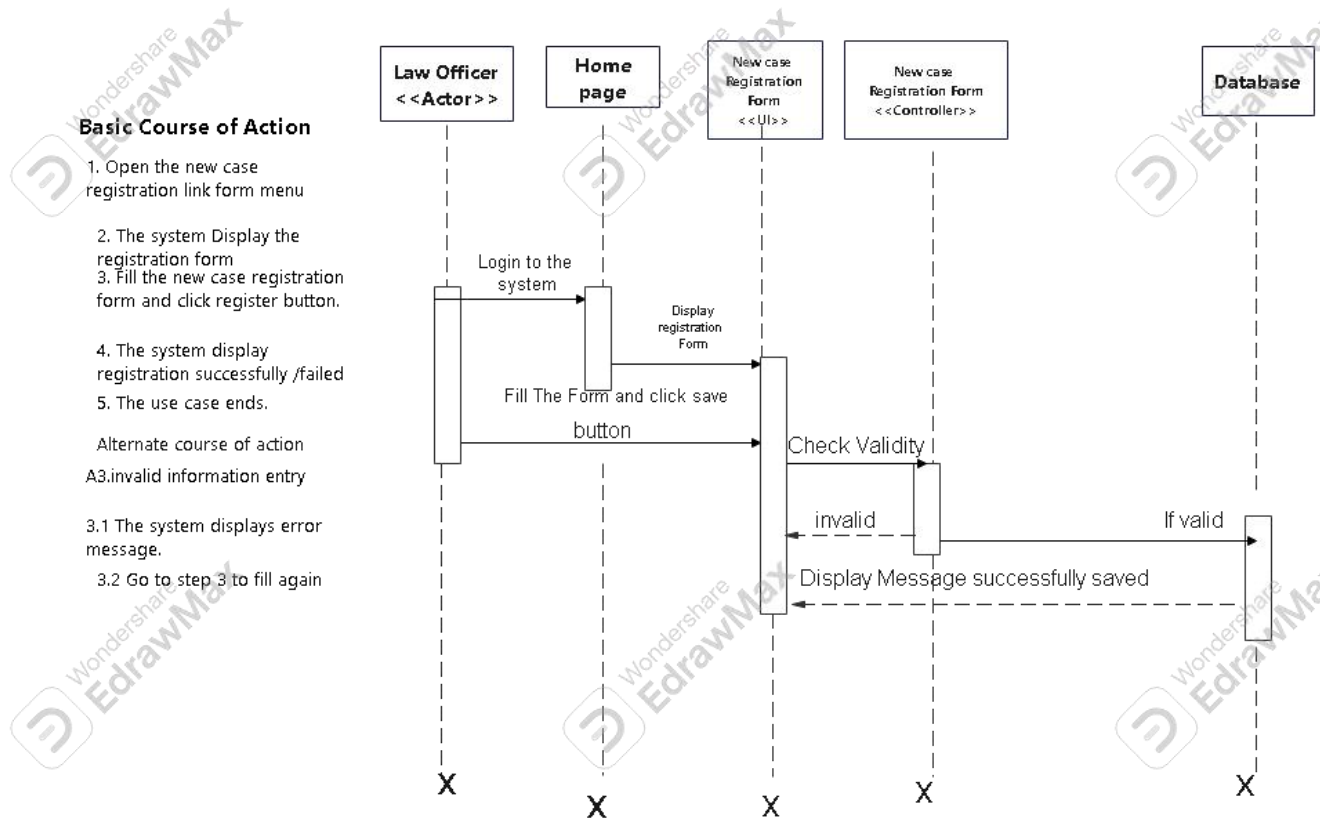


Figure 3: Sequence diagram for new case registration

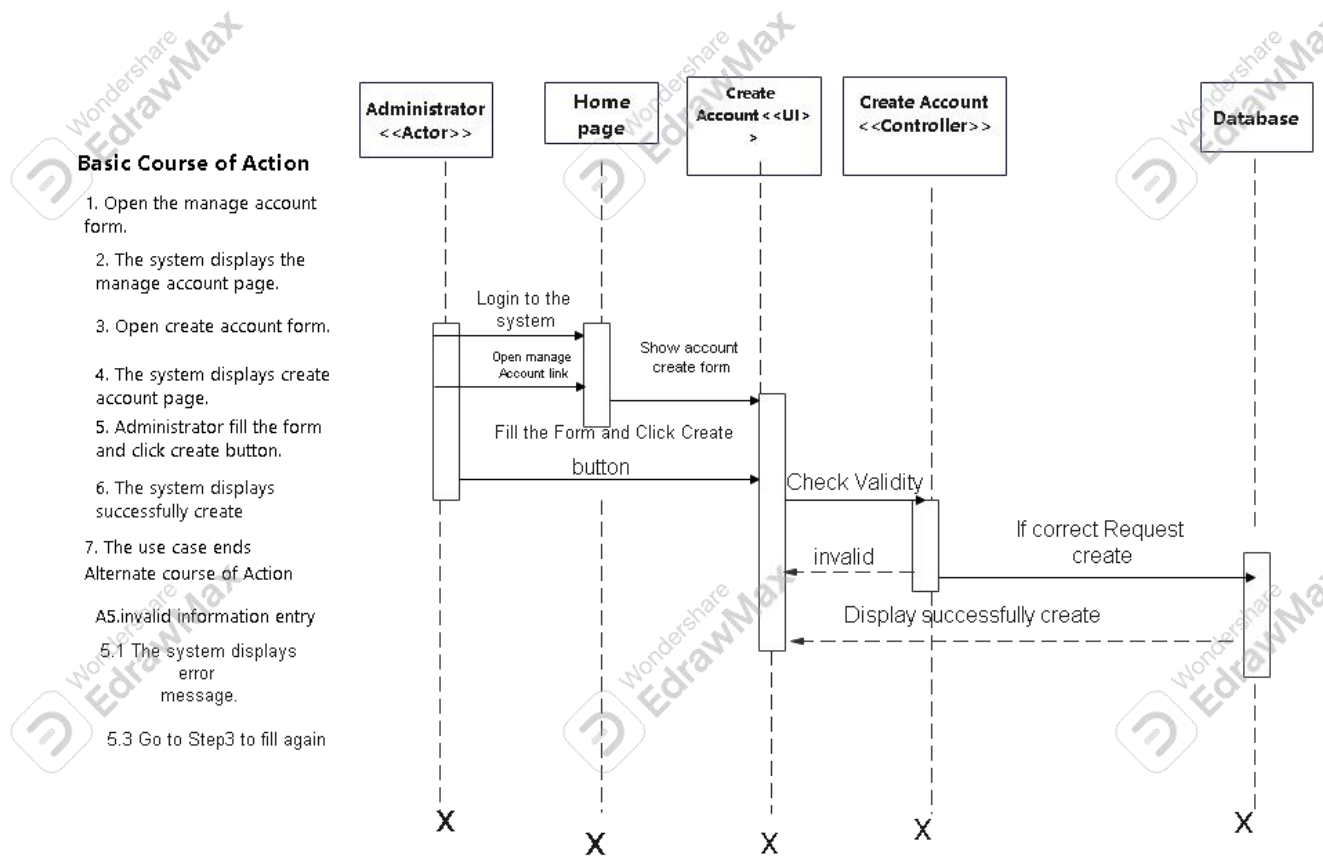


Figure 4: Sequence diagram for create account

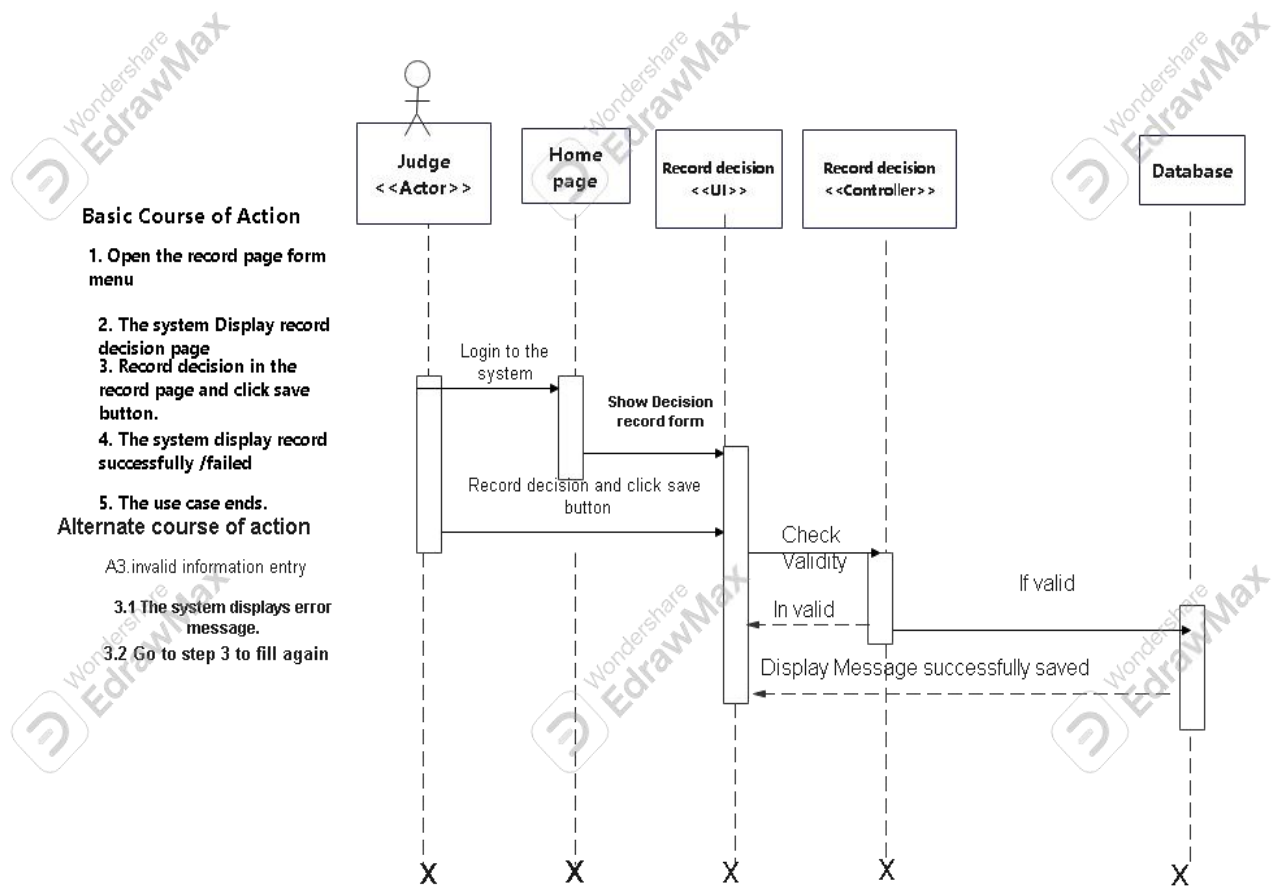


Figure 5: Sequence diagram for record decision

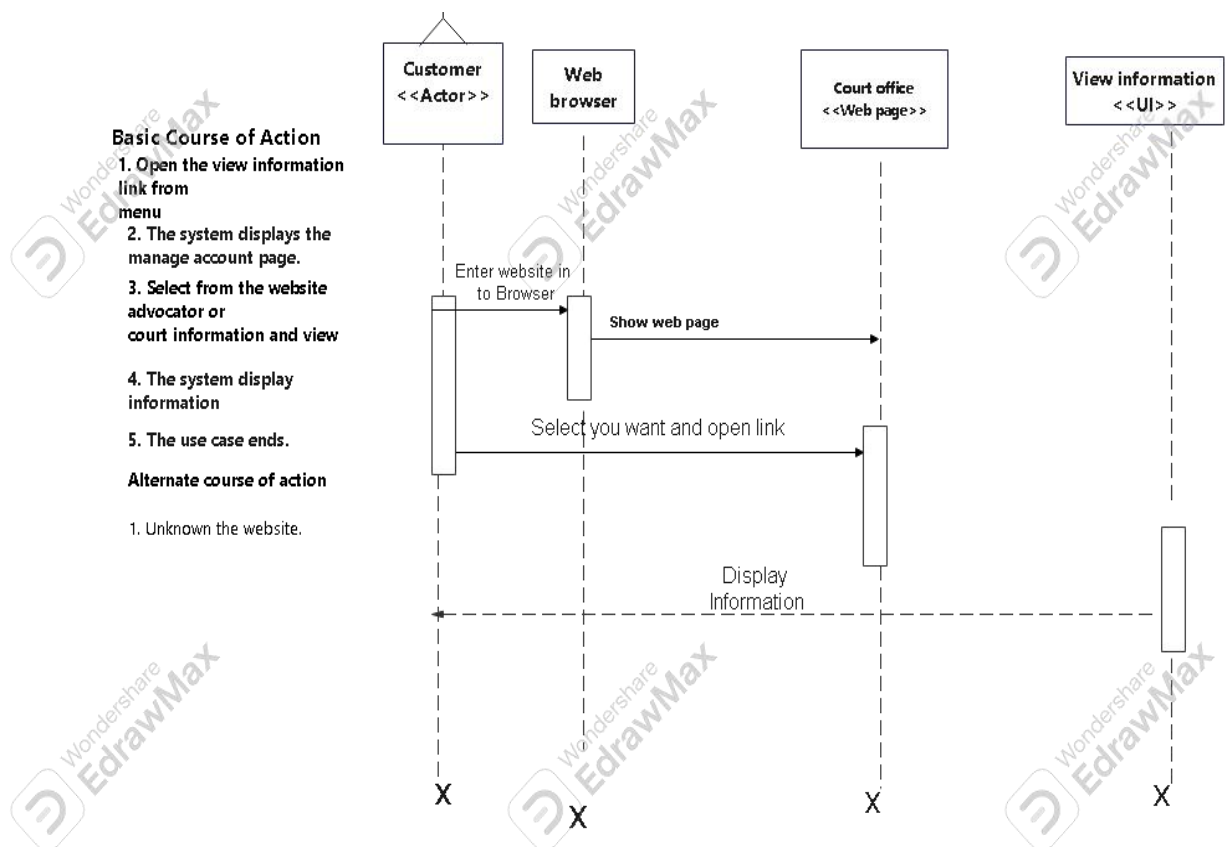


Figure 6: Sequence diagram for view information

4.5 Activity diagrams

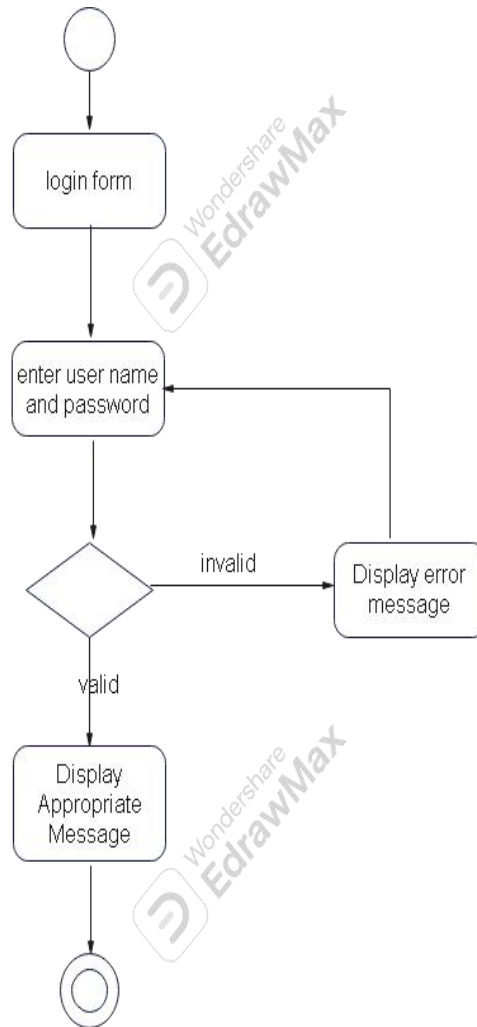


Figure 7: Activity Diagram for Login

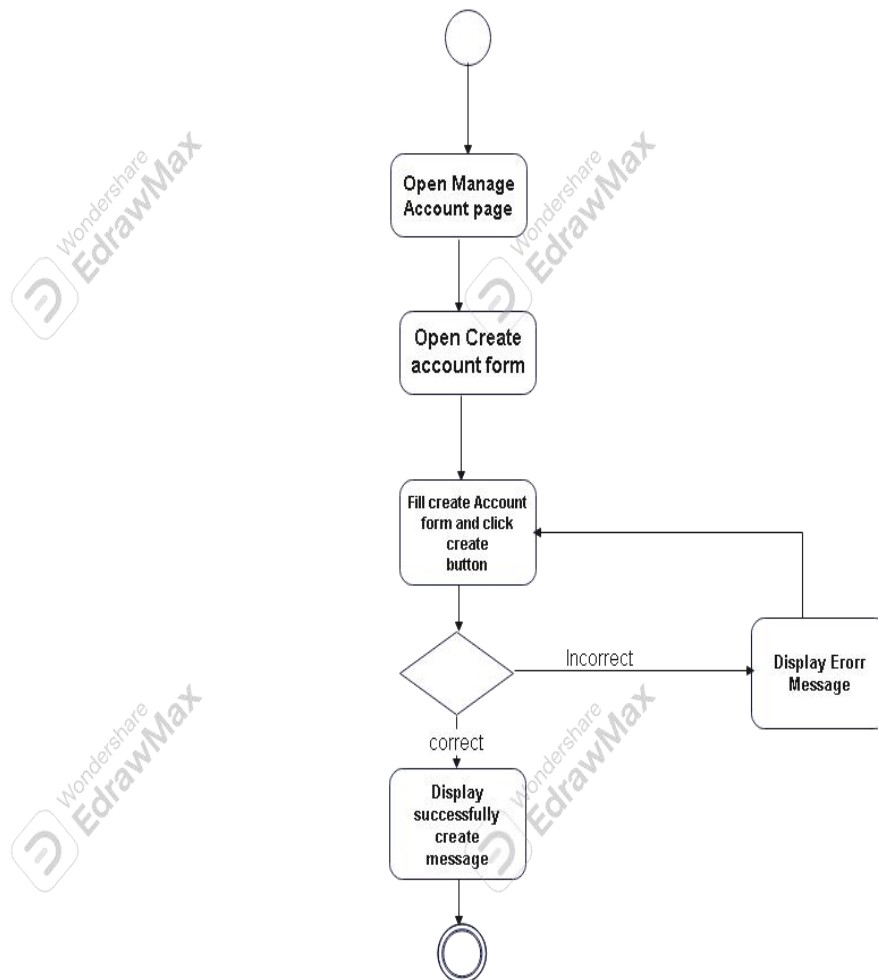


Figure 8: Activity Diagram for create account

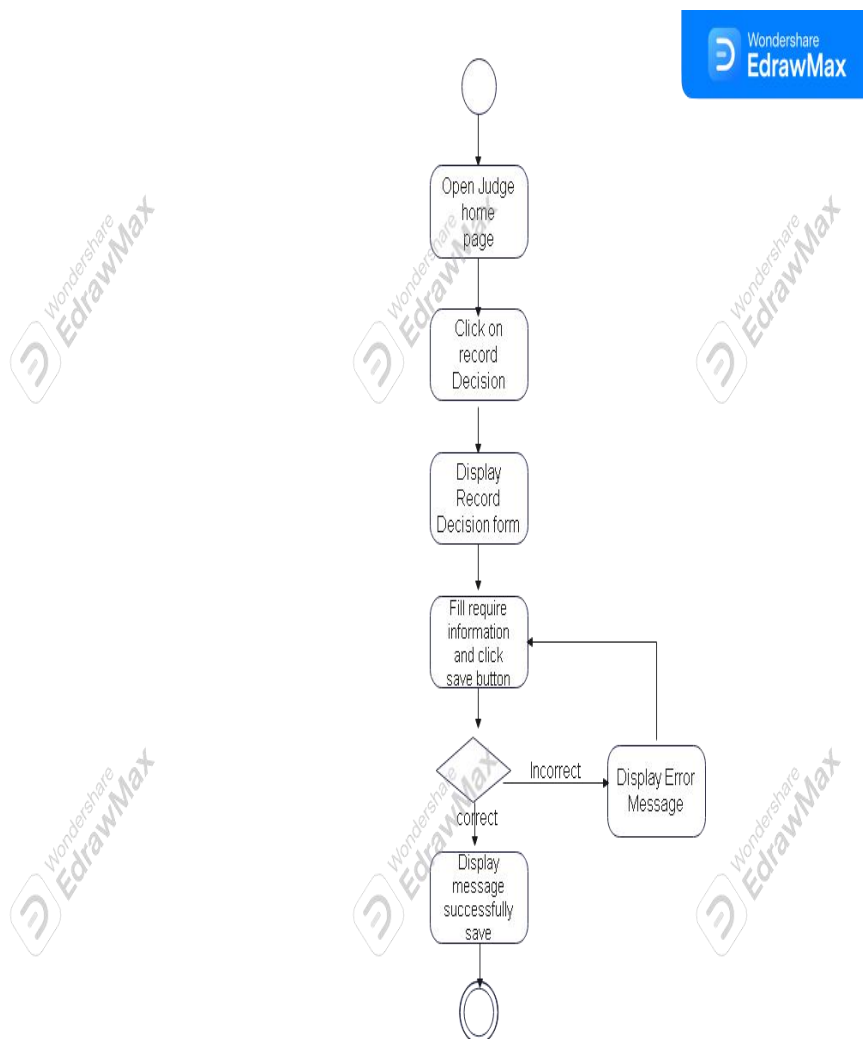


Figure 9: Activity Diagram for record decision form

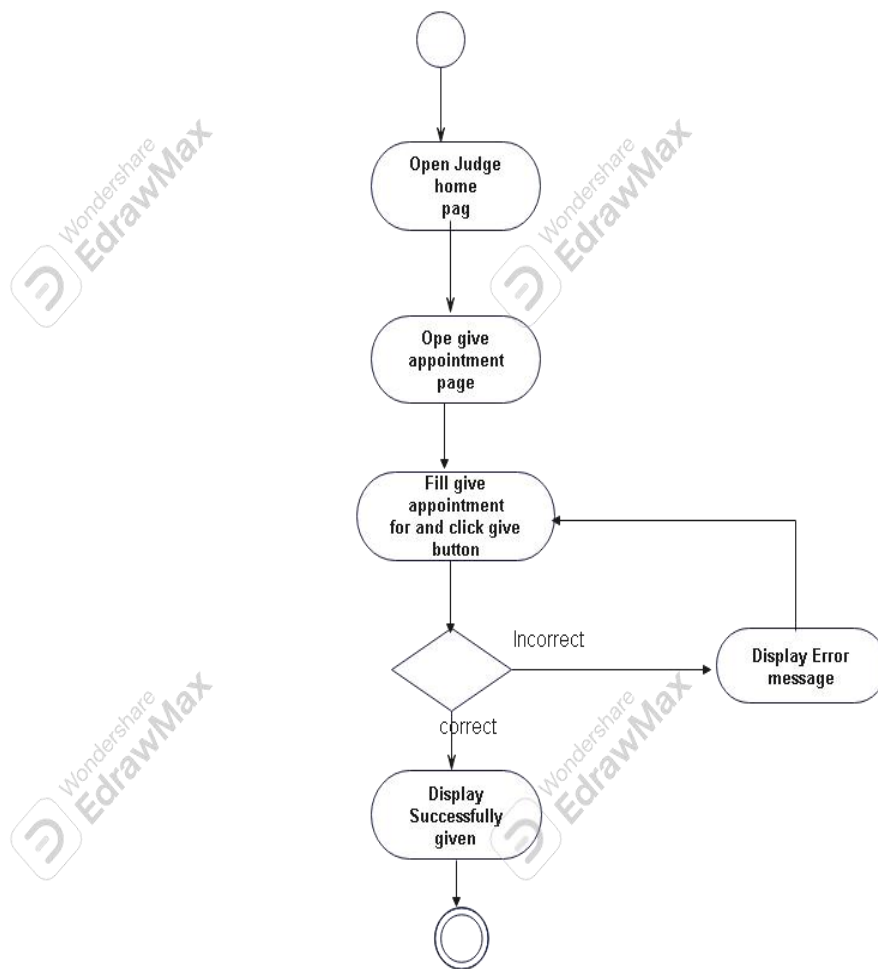


Figure 10: Activity Diagram for appointment page

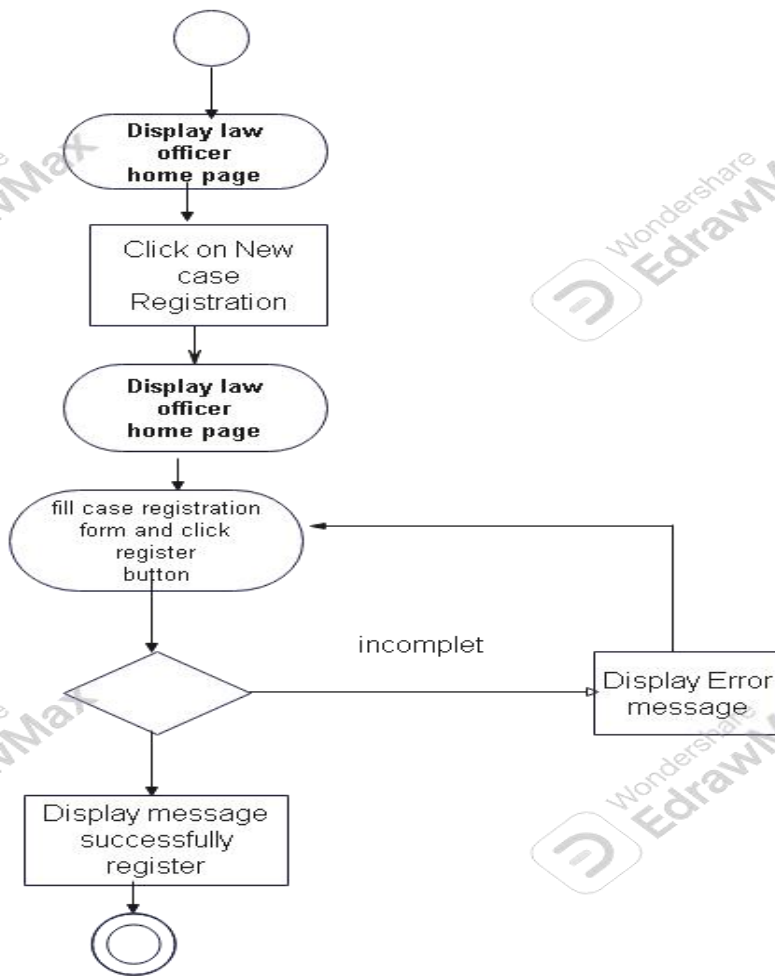


Figure 11: Activity Diagram of officer home page

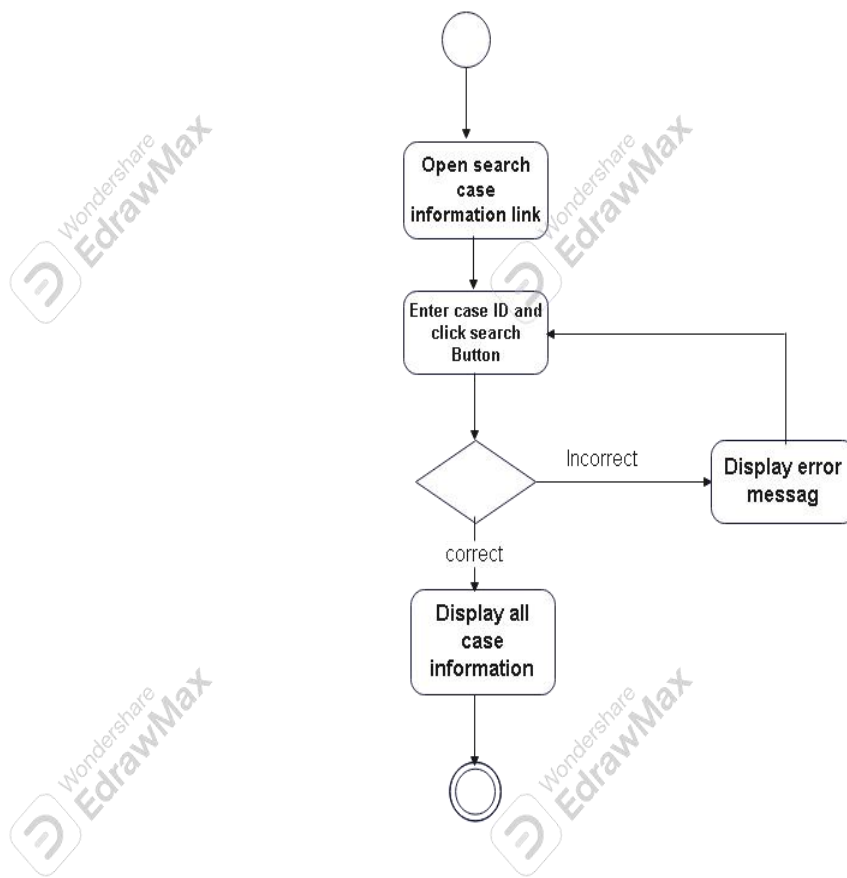


Figure 12: Activity Diagram for search case information

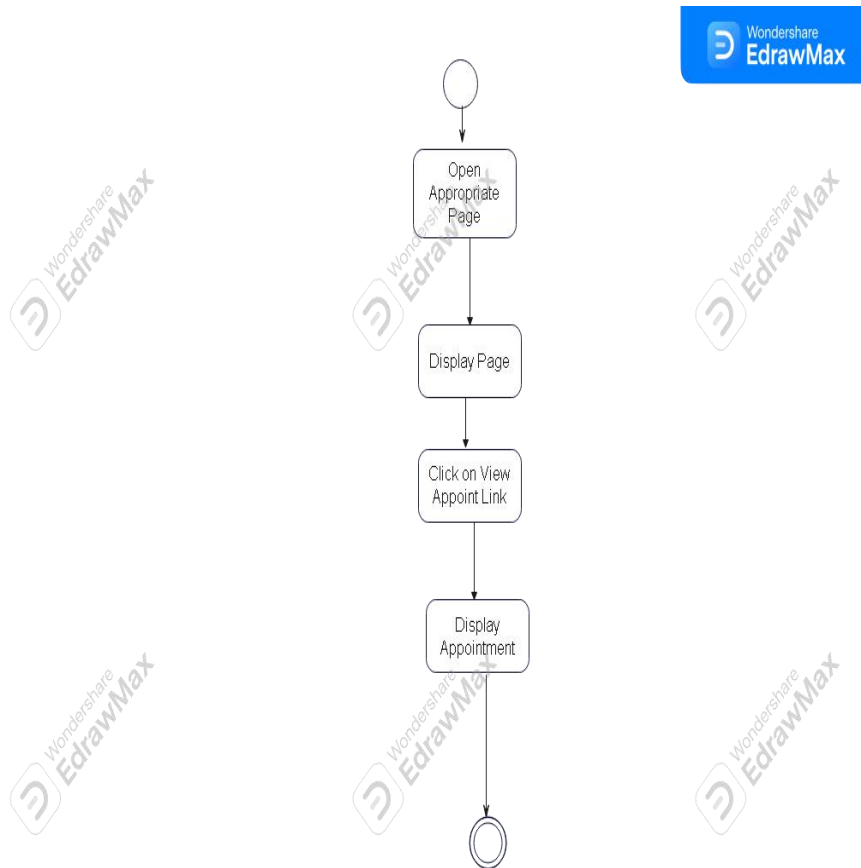


Figure 13: Activity Diagram for view appointment

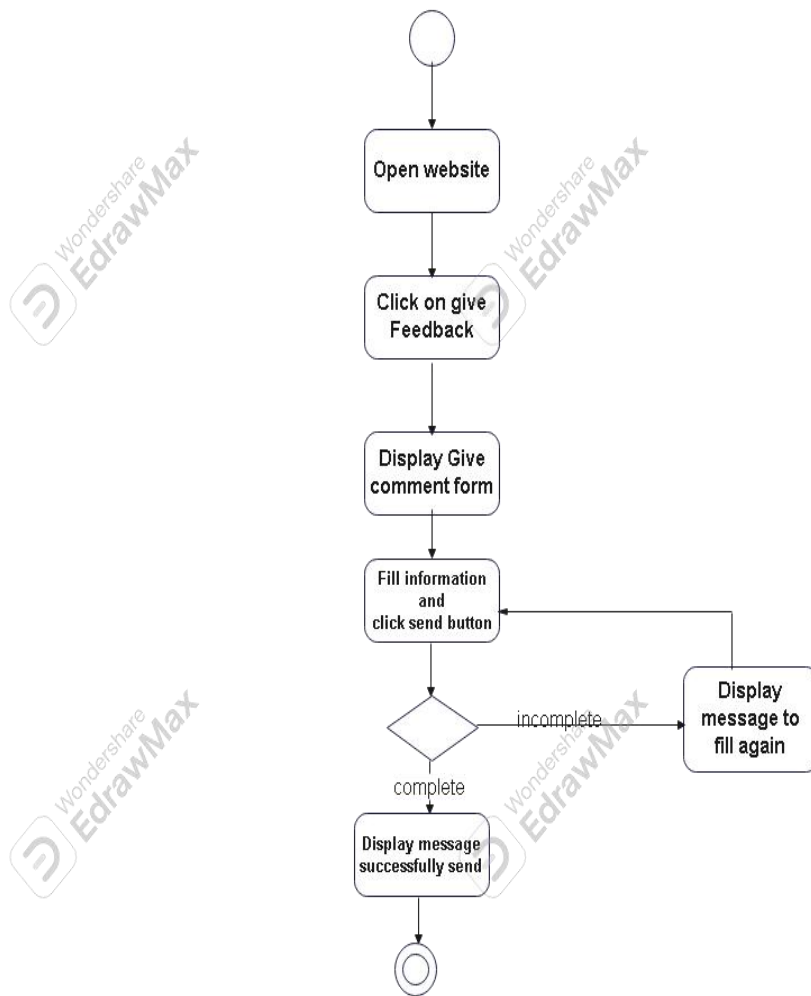


Figure 14: Activity Diagram for feedback

4.6 State Chart Diagram

State diagrams model the system from the point of view of a single class and the events that can affect the objects of the class. They show all possible behaviors of objects of a class and record the ordering of events[4].

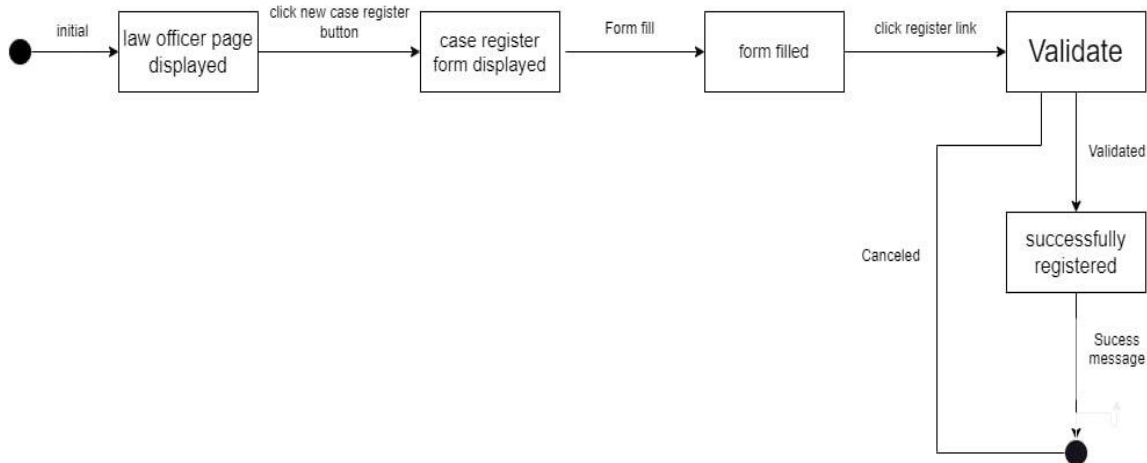


Figure 16: state chart New case registration

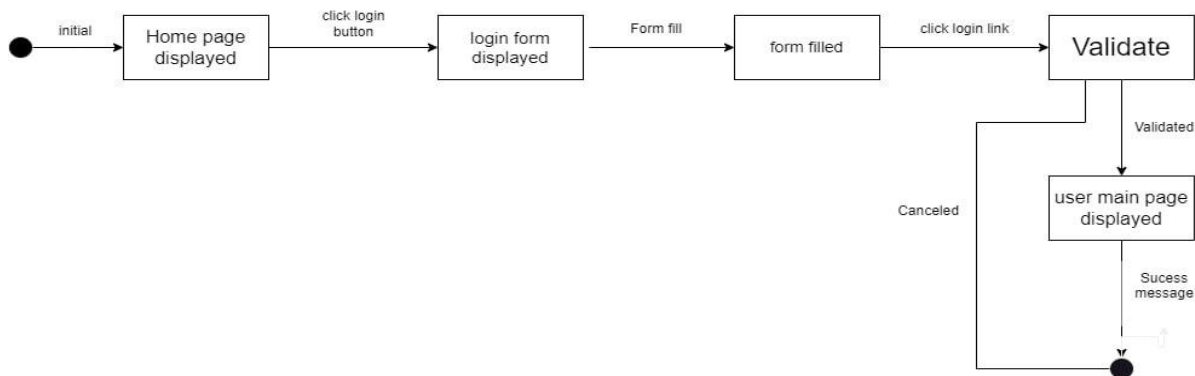
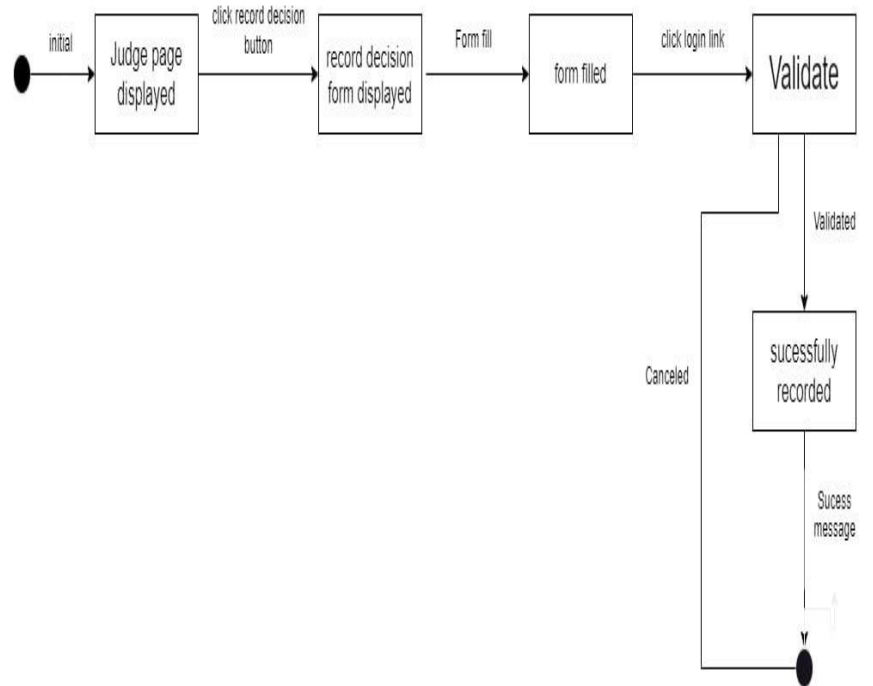


Figure 17: state chart login page



Figer 18:state chart record desion

4.7 Class diagram

The class diagram is the main building block of object oriented modeling. It is used both for general conceptual modeling of the systematic of the application, and for detailed modeling translating the models into programming code. The classes in a class diagram represent both the main objects, interactions in the application and the classes to be programmed[3]. In the class diagram, classes are represented with boxes which contain three parts:

The top part contains the name of the class

The middle part contains the attributes of the class

The bottom part gives the methods or operations the class can take or undertake

The main action that shows on class diagram is class, object, operation, and dependencies.

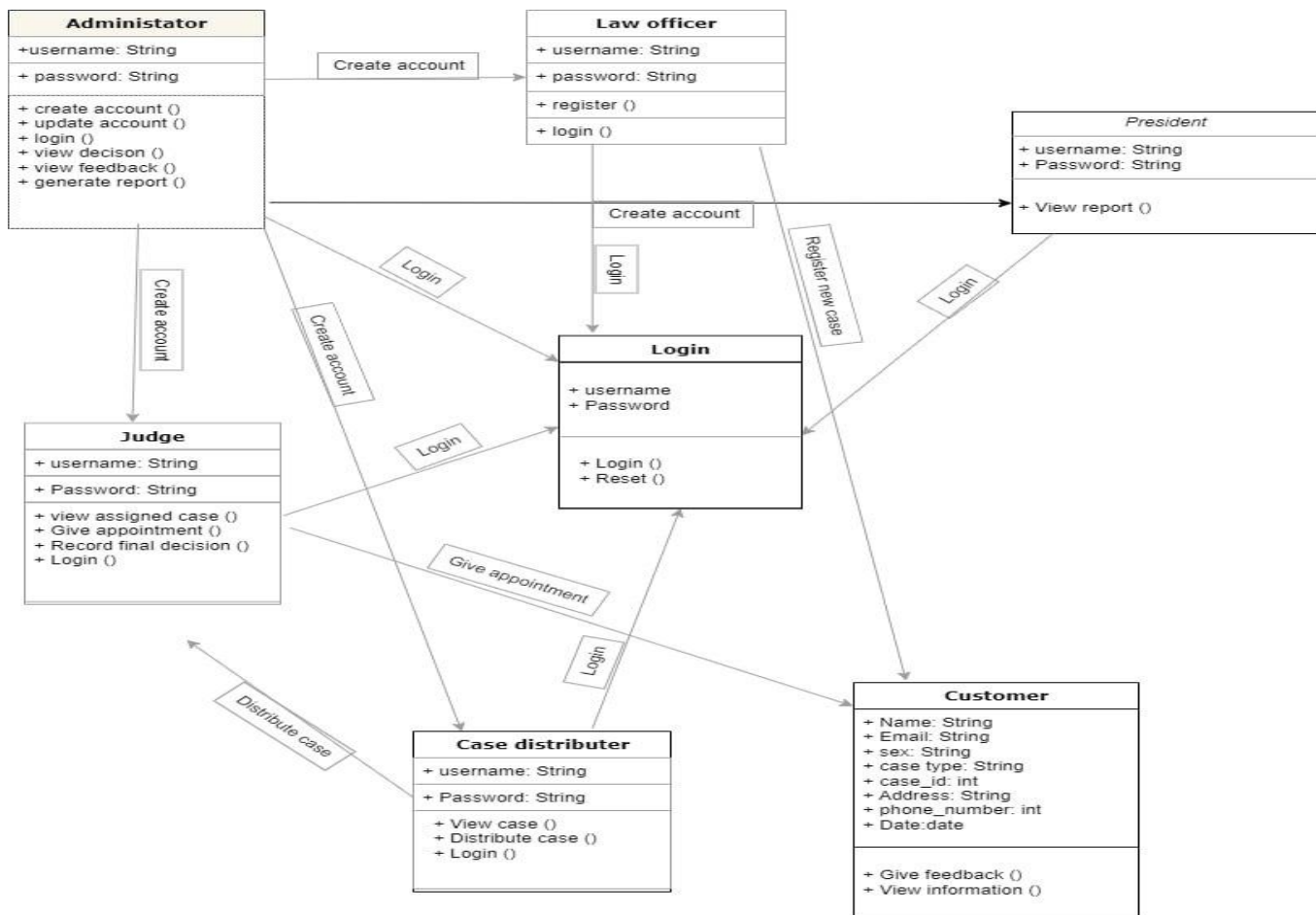
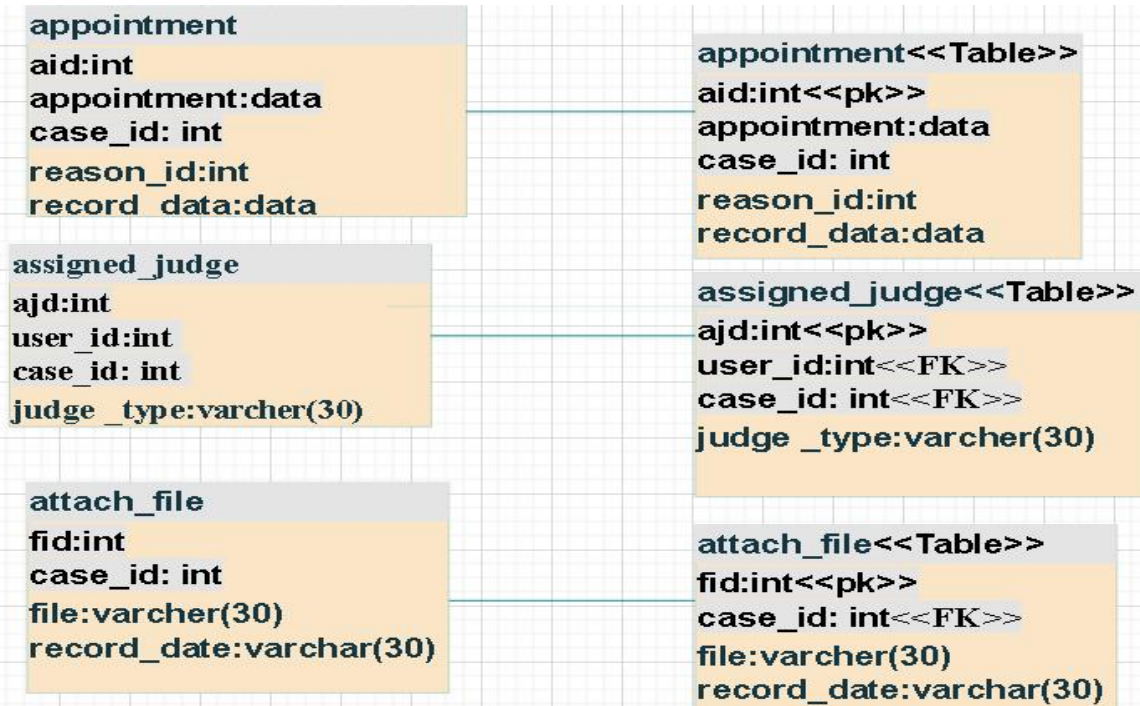


Figure 21: Class diagram

4.9 Persistent modeling/database design

Database design is the process of producing a detailed data model of a database. This data model contains all the needed logical and physical design choices and physical storage parameters needed to generate a design. The term database design could also be used to apply to the overall process of designing, not just the base data structures, but also the forms and queries used as part of the overall database application within the DBM[5]



```
case
cid:int
case_id: int
plaintiff:varcher(30)
defendent:varchar(30)
case_type : int
decision :varchar(30)
case_status varchar(33)
```

```
case<<Table>>
cid:int<<pk>>
case_id: int<<FK>>
plaintiff:varcher(30)
defendent:varchar(30)
case_type : int
decision :varchar(30)
case_status varchar(33)
```

```
case_info
kid:int
case_id: int
first_name:varcher(30)
father_name:varchar(30)
grandfather_name:varchar(30)
gender:varchar(30)
email:varchar(30)
region :varchar(30)
zone:varchar(33)
wereda:varchar(30)
kebele:varchar(30)
wegen:varchar(30)
argumer_money:varchar(30)
judgement_money:varchar(30)
litiget_type:varchar(30)
phone:varchar(30)
file_page:int
file:varchar(30)
```

```
case_info<<Table>>
kid:int<<pk>>
case_id: int<<FK>>
first_name:varchar(30)
father_name:varchar(30)
grandfather_name:varchar(30)
gender:varchar(30)
email:varchar(30)
region :varchar(30)
zone:varchar(33)
wereda:varchar(30)
kebele:varchar(30)
wegen:varchar(30)
argument_money:varchar(30)
judgement_money:varchar(30)
litiget_type:varchar(30)
phone:varchar(30)
file_page:int
file:varchar(30)
```

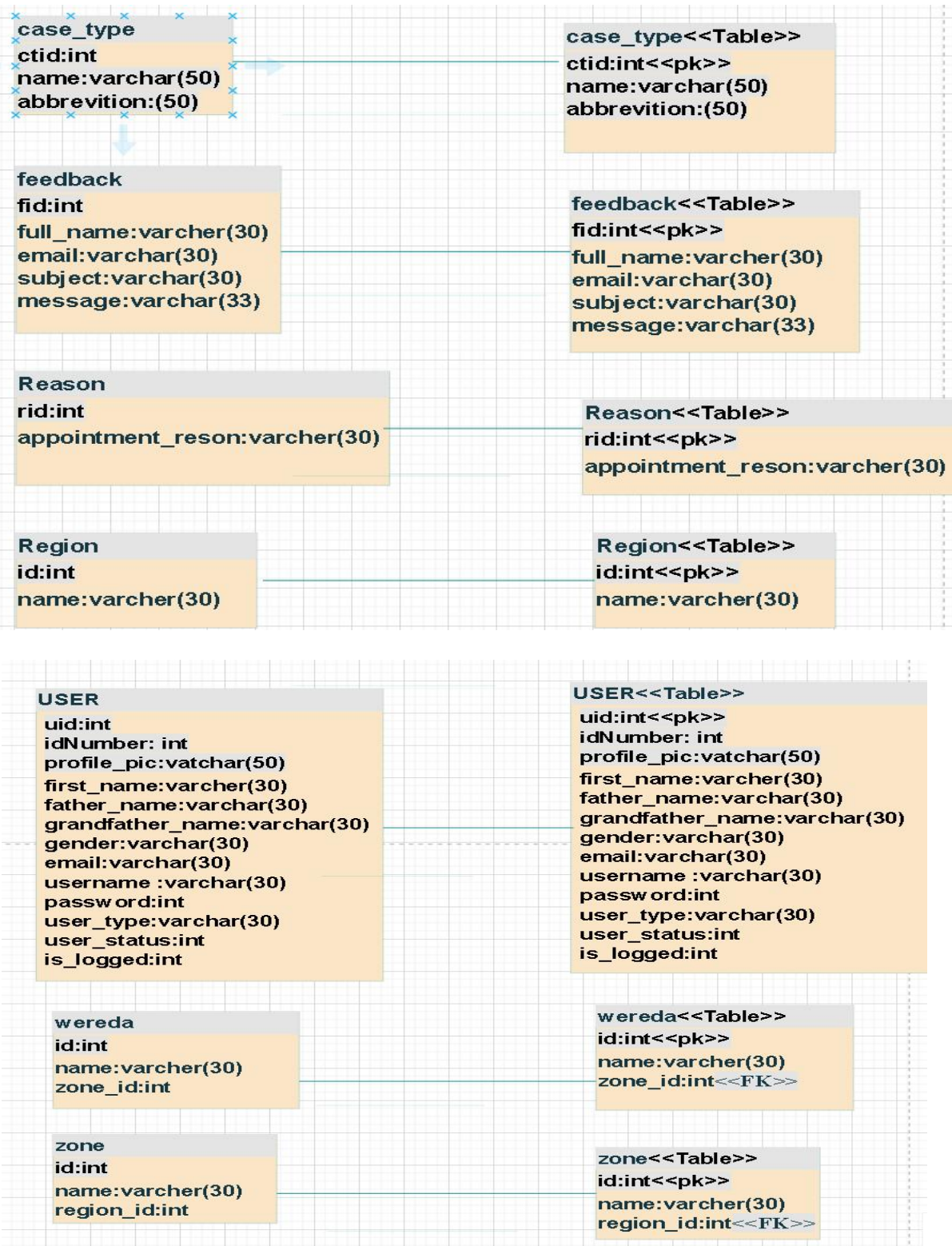



Figure 22 : Persistent modeling

4.10 Component Diagram

Component diagrams are used in modeling the physical aspects of object-oriented systems that are used for visualizing, specifying, and documenting component-based systems. Component diagrams are essentially class diagrams that focus on a system's components that often used to Model the static implementation view of a system[6].

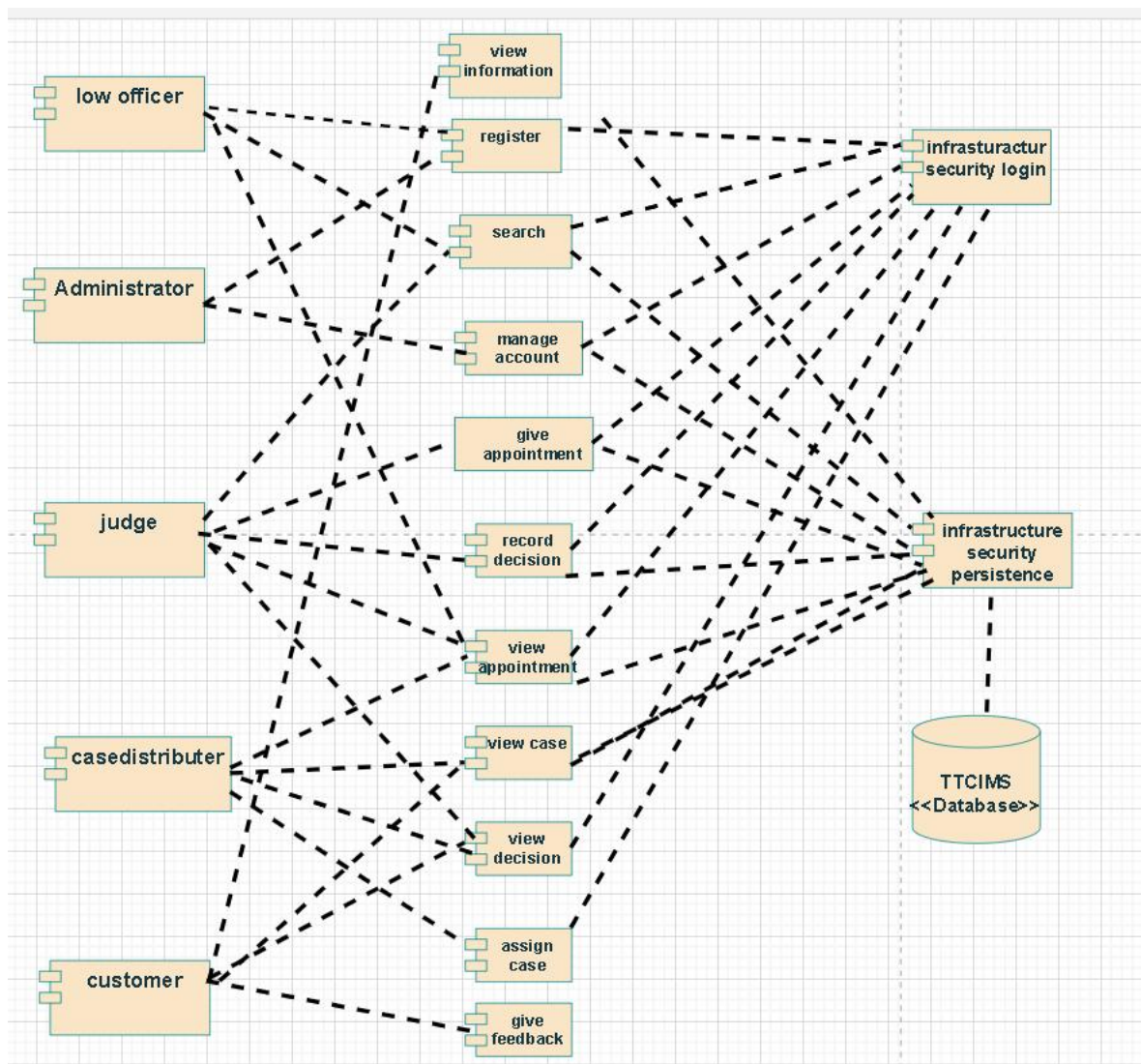


Figure 23 : Component diagram

4.11 Deployment Diagram

A UML deployment diagram is a diagram that shows the configuration of run time processing nodes and the components that live on them. A deployment diagram is a kind of structural diagram used in modeling the physical aspects of an object-oriented system. They are often be used to model the static deployment view of a system [7].

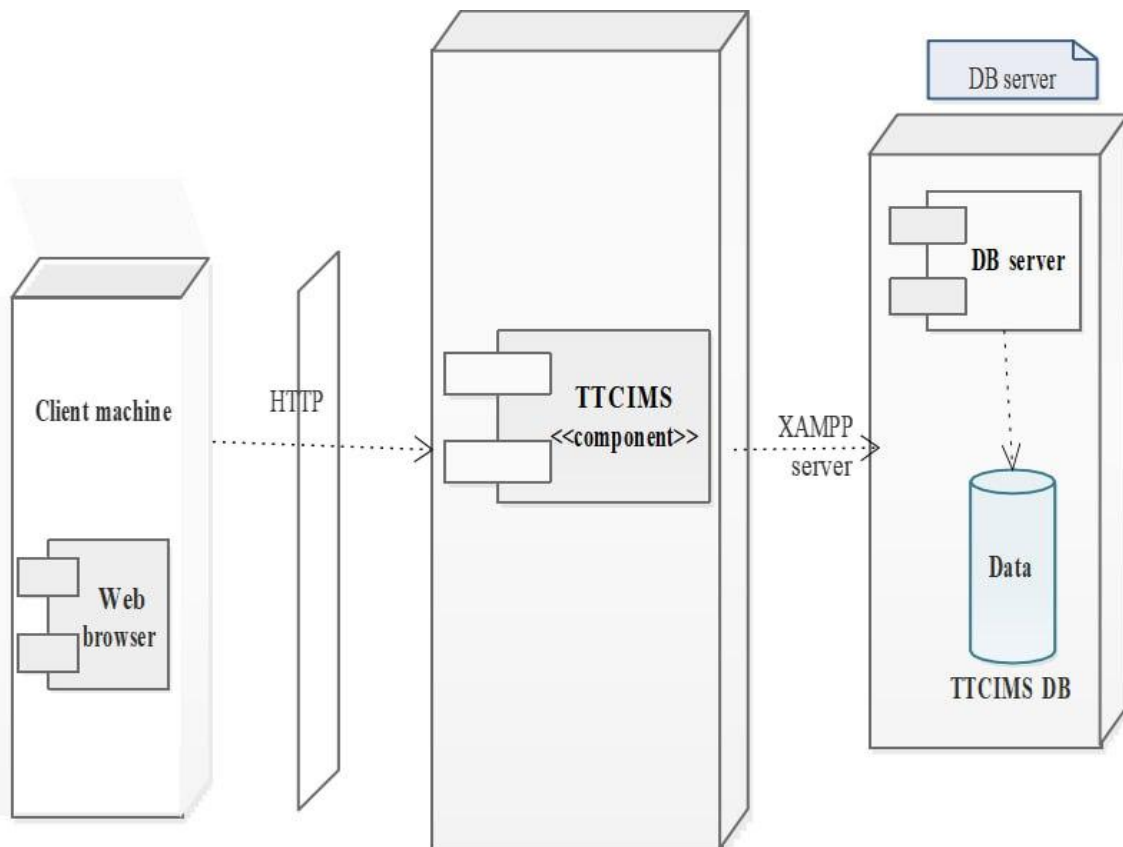


Figure 24 : deployment diagram

CHAPTER FIVE

SYSTEM IMPLEMENTATION

5.1 Introduction

The implementation phase is the most crucial phase in which it transforms the design and analysis of the system into a tangible system by writing the code to the system to be developed and make it operational and applicable by testing and debugging the functionalities that are done. This phase involves the construction of the actual project result. During this phase that the project becomes visible to outsiders, to whom it may appear that the project has just begun. This makes the implementation stage more essential step to develop the required system. So, it is the most vital and necessary stage in achieving a successful system and in giving the users confidence that the new system will work and be effective by testing the system that is already implemented. In this phase, the production system is installed, initial user training is completed, user documentation is delivered, and the post implementation review meeting is held. When this phase is completed, the application is in steady state production. Once the system is in steady-state production, it is reviewed to ensure that we met all of the goals in the project plan for a satisfactory result. The result of this phase consists of source code, together with documentation to make the code more readable. This is what we call software implementation. The purpose of these activities is to convert the final physical system specification into working model with reliable software and hardware, document the work that has been done, and provide help for current and future users and take care of the system.

5.2 Sample code

```
<?php
include '../CIMS/login/loginHeader.php';
?>
<?php
$username = $password = "";
$username_err = $password_err = $all_err = "";
$test = true;

if (isset($_POST["login"]) and ($_SERVER["REQUEST_METHOD"] ==
"POST")) {
    // Validate username
    if (empty($_POST["username"])) {
        $username_err = "Please enter your username";
        $test = false;
    } else if (validateIdNumber($_POST["username"]) == 0) {
        $username_err = "Please enter valid username";
        $test = false;
    } else {
        $username = $_POST["username"];
    }
    // Validate password
    if (empty($_POST["password"])) {
        $password_err = "Please enter your password";
        $test = false;
    } else if (validateIdNumber($_POST["password"]) == 0) {
        $password_err = "Please enter valid password";
        $test = false;
    } else {
```

```
$password = $_POST["password"];
}
// Check if the user is already logged in
// if (checkIs_loggedIn(1)) {
    // $all_err = "You are already login.";
// } else {
// Check user credentials
if ($test == true) {
    if (checkUserByUsername($username)) {
        $user_data = checkUserCredentials($username , $password);
        if ($user_data) {
            if ($user_data['user_status'] == 2) {
                $all_err = "This account is deactivated. Please contact the admin.";
            } else if ($user_data['user_status'] == 0 || $user_data['user_status'] == 1)
        {

            $_SESSION["uid"] = $user_data['uid'];
            updateUserStatus(1, $_SESSION["uid"]);
            // updateIs_loggedIn(1, $_SESSION["uid"]);
            if ($user_data['user_type'] == "Admin") {
                header('location: Admin/admin.php');
            } else if ($user_data['user_type'] == "Law_officer") {
                header('location: law_officer/law_officer.php');
            } else if ($user_data['user_type'] == "Judge") {
                header('location: judge/judge.php');
            } else if ($user_data['user_type'] == "Case_distributor") {
                header('location: case_distributor/case_distributor.php');
            } else if ($user_data['user_type'] == "President") {
                header('location: president/president.php');
            }
        }
    }
}
```

```
        }
    } else {
        $all_err = "There is no user associated with the given information";
    }
} else {
    $all_err = "There is no user associated with the given information";
}
}
}
//}
?>

<!-- Navigation Bar -->
<nav class="navbar navbar-expand-lg navbar-dark bg-dark mb-4">
    <div class="container">
        <a class="navbar-brand fw-bold" href="#">Yeki Woreda Court</a>
        <button class="navbar-toggler" type="button" data-bs-toggle="collapse"
data-bs-target="#navbarNav">
            <span class="navbar-toggler-icon"></span>
        </button>
    </div>
</nav>

<div class="container d-flex justify-content-center align-items-center min-vh-
100">
    <div class="card shadow-lg rounded-4 p-4" style="max-width: 420px; width:
100%;">
        <div class="card-body">
            <div class="text-center mb-4">
                <h2 class="fw-bold mb-1">Login</h2>
            </div>
```

```
<?php if (!empty($all_err)) { ?>
    <div class="alert alert-danger fw-bold text-center">
        <?php echo $all_err; ?>
    </div>
<?php } ?>
<form action="" method="POST" novalidate>
    <div class="mb-3">
        <label for="username" class="form-label fw-
semibold">Username</label>
        <input type="text" class="form-control <?php
echo !empty($username_err) ? 'is-invalid' : ''; ?>" id="username"
name="username" value="<?php echo $username; ?>">
        <div class="invalid-feedback">
            <?php echo $username_err; ?>
        </div>
    </div>
    <div class="mb-3">
        <label for="password" class="form-label fw-semibold">Password</label>
```

5.3 User interface

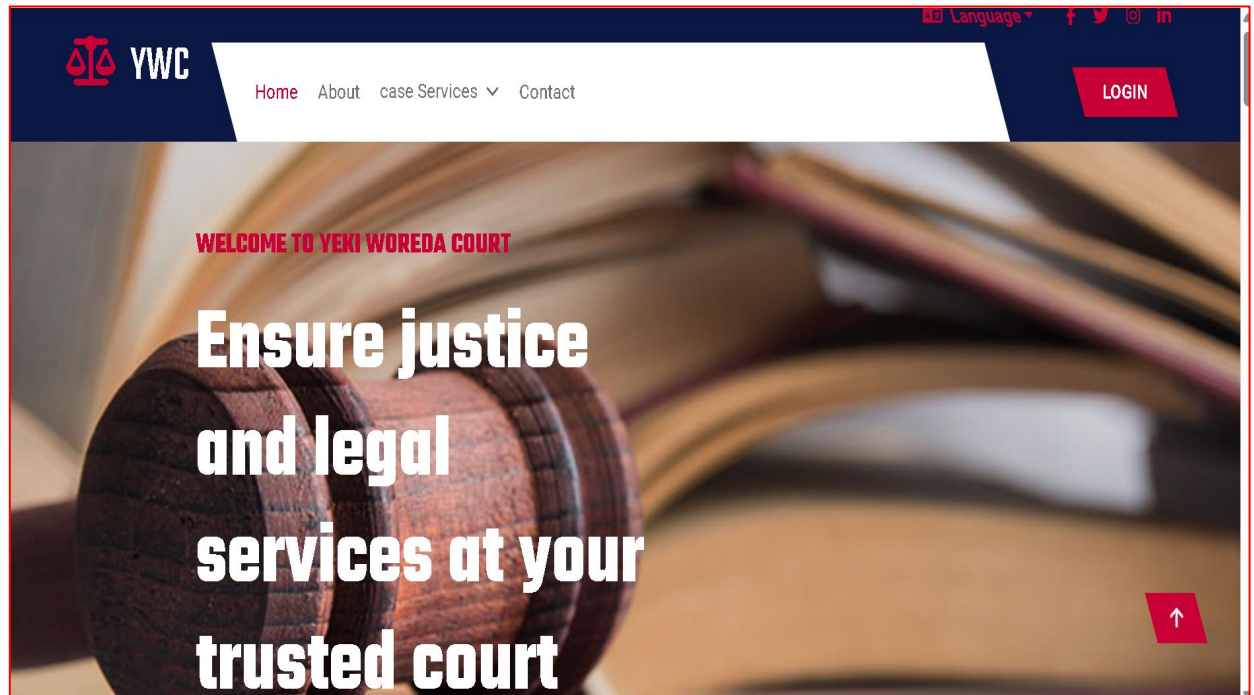
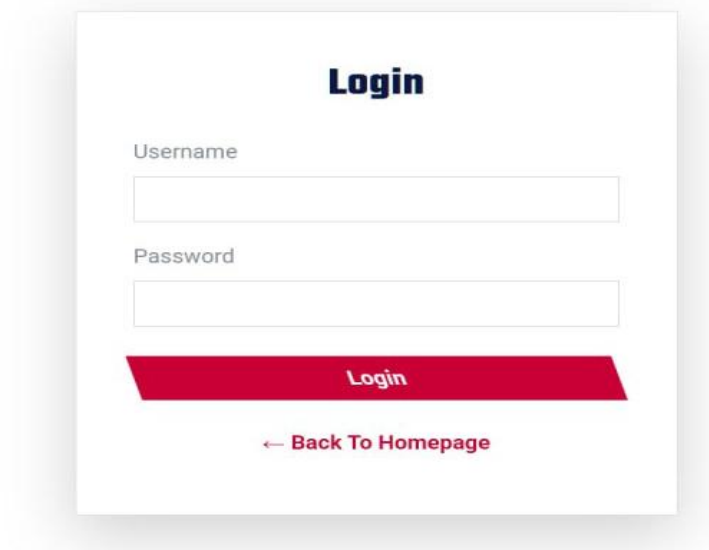
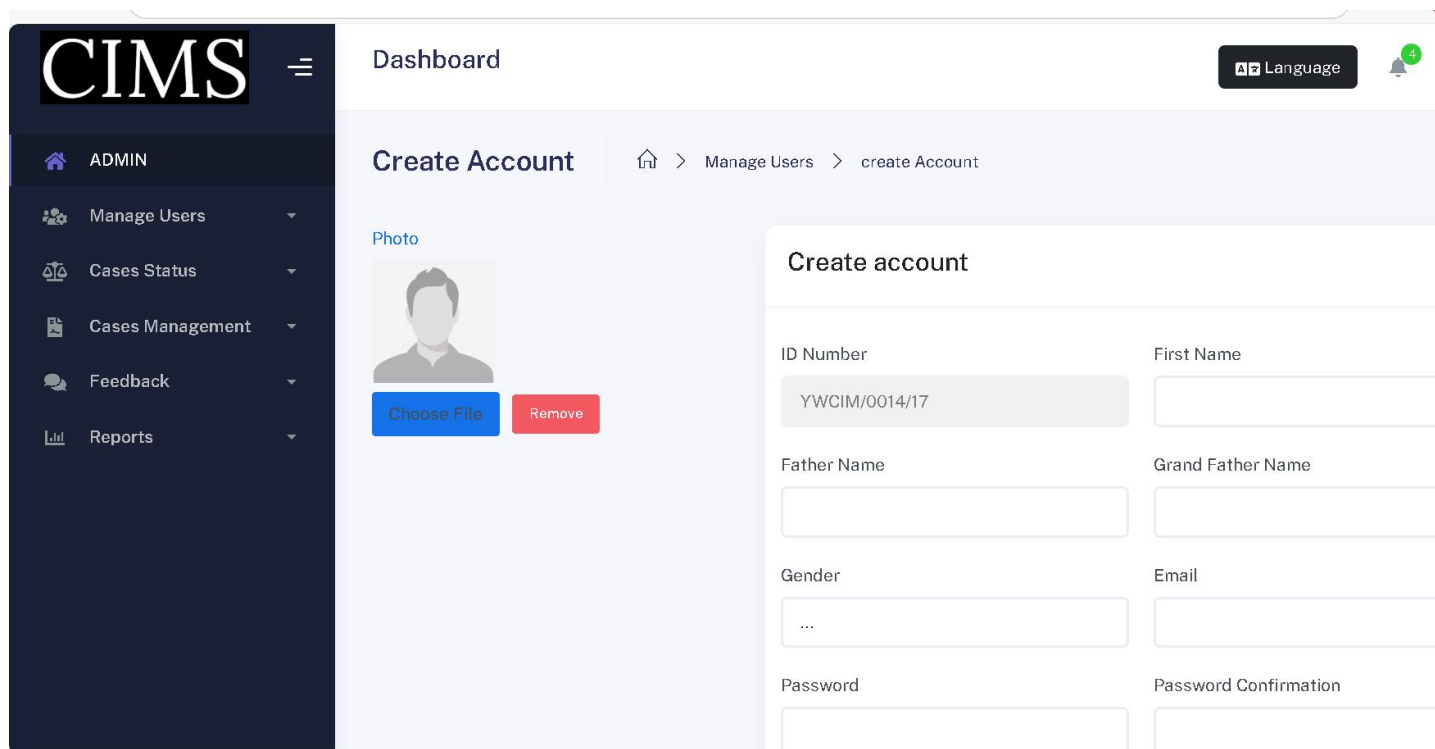


Figure 26: user interface home page



The login page features a white card with a red header titled "Login". Below the header are two input fields: "Username" and "Password". A red "Login" button is positioned below the password field. At the bottom of the card, there is a red link that says "← Back To Homepage".

Figure 24: user interface login page



The "Create Account" page is part of the CIMS dashboard. It has a dark sidebar with the CIMS logo and a menu including ADMIN, Manage Users, Cases Status, Cases Management, Feedback, and Reports. The main content area is titled "Create Account" and includes a breadcrumb trail: Home > Manage Users > create Account. On the left, there is a "Photo" section with a placeholder image and buttons for "Choose File" and "Remove". On the right, the "Create account" form contains several input fields: ID Number (pre-filled with YWCIM/0014/17), First Name, Father Name, Grand Father Name, Gender (with a dropdown menu), Email, Password, and Password Confirmation. The top right of the dashboard shows a "Language" dropdown and a notification bell icon with a count of 4.

Figure 25: user interface creat account page

5.4 Testing

The test plan is designed to analyze the logic used in the implementation of the court information system. The tests will allow us to ensure correct program flow, and to determine the error-handling capability of the system. Once coding has begun, the testing process can be beginning and proceed in parallel. As each program module is produced, it can be tested individually, as a part of larger program, and then as a part of larger system.

Unit testing: Unit test is a way of testing each of the system functionality independently.

Accordingly, the team has tested each one of the major activities and the rest accompanying activities independently using different user input, different login mechanisms and any technique of fault finding so that an incorrect functioning of the activities was corrected at the right time.

Integration Testing: Integration testing is testing in which a group of components are combined to produce output. Also, the interaction between software and hardware is tested in integration testing if software and hardware components have any relation.

System testing: It is the final step of testing. In this system tested the entire system as a whole with all forms, code, modules. In this we tested all the functionalities in the System. All errors in the forms, functions, modules have been tested. Finally, System testing ensures that the entire integrated software system meets the desired requirements. It tests a configuration to ensure known and predictable results.

5.5 User manual preparation

User manual preparation since the system is web based application everything important for the user will be explained and implemented while preparation of short training document when the system is deployed.

5.6 Training

Training manuals serve the important purpose of providing a consistent way to communicate instructions to employees about how to perform essential functions of their jobs by using the system. To be effective, training manuals is based on functionalities and instructional objectives. Information should be provided about the processes and procedures employees are required to follow as well as tasks that form the basis of the jobs they are charged with performing.

5.7 Start-up strategy

The start-up strategy will be:

Buying a domain name from a company with its corresponding IP address for the server.

Hosting the system. Finally, available internet connection for users.

5.8 Conclusion and recommendation

5.8.1 Conclusion

When we conclude about our system this system has five phases these are Proposal, Analysis, Design, Implementation and Conclusion.

The analysis phase helps the team to well understand the major Functional and Nonfunctional requirements, use case, sequence, and activity diagrams of the system. And also we evaluate the existing system weakness and strength. After understand the existing manual system we gathering information from different source and analysis, design, implement it to solve the problem of TTCIMS. From this we perform keeping the data of customer, customers can access his/ her data by using their own username and password like view appointment view decision and also for better communication between the employee. Case registration, Give appointment, record decision, create account and generate report.

Generally our group team have pass different activities to solve the problem TTCIMS like time wastage, data redundancy and lack of security.

5.8.2 Recommendation

The team would recommend that further work should be done on the system. Based on the preceding information in our project, we believe that the following recommendation will help the organization to improve their system more reliable and available as they want. The organization should be inter connect with zonal court Office communicate online. The team recommend that it's better to be integrated with expert system.

Finally the team would recommend that future work should add video conference on the system.

Reference

- [1] h. I. Jacobson, Use case modeling, 2nd ed., jacky, Ed., Spen: Wesley, 2003.
- [2] online, what is class modeling, 2nd update ed., james, Ed., usa: adventure works press, 27 September 2021.
- [3] <http://Wikipedia>, sequence diagram, 2nd ed., james, Ed., america: perisedant, 2006
- [4] Lazarevska, what is persistence models, 2nd ed., Syed, Ed., asia : worker press, 2016.
- [5] J. Rumbaugh, what is component diagram, 2nd ed., G. Booch, Ed., germany: creator, 1994.
- [6] online, deployment, 2nd ed., chen, Ed., asia: the deployment diagram describes the physical deployment of information generated by the software program on hardware components., 2008.
- [7] L. E, Business Judgment Rule, 2nd ed., chen, Ed., america: business associations, 2009.
- [8] T. A, the Business Rule, 2nd ed., jacky, Ed., america: adventure works , 2000

Appendix

- ❖ Here are the lists of questions that we were used to interview and data gathering with the Tepi town yeki wereda court office.
 1. how many sectors are there according to the current office strategies and work situation?
 - 2, How the sectors and employees work their activity?
 3. how we could to get any manual data from the information store for example Annual schedule, List of sectors etc.
 4. Who schedule the annual and semi-annual schedule?
 5. What is the role of law officer, judge, advocator public proctor client president?
 6. When the report is generated?
 7. How much max duration of one case to complete?
 8. How does your current system work?
 9. How you make relationship with the user?
 10. What are the procedures or steps when customers get service of the organization?
 11. How do you store customer's data and other information?
 12. What are the problems that you face when the users use the existed system?